

# An Excel-Based Companion to



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This document and the associated data files and spreadsheets can be found at <http://brucehardie.com/notes/047/>.

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# Chapter 1

## Introduction

A customer-base audit is a systematic review of the buying behavior of a firm's customers using data captured by its transaction systems. The objective of this audit is to provide an understanding of how customers differ in their buying behavior and how their buying behavior evolves over time.

In *The Customer-Base Audit* (hereafter *TCBA*), we present the core analyses we associate with a customer-base audit, grouping them according to five “lenses”. In this companion, we show how to create almost all of the Lenses 1–5 figures and tables presented in the book using Excel.<sup>1</sup> We will ignore any analyses that take the product dimension into account (*TCBA*, Chapter 8), as undertaking them in Excel is quite cumbersome.

This note is written with three audiences in mind:

- Executives who see the value of the analyses presented in *TCBA* and have someone who can do the analyses for them, but who would like to have a deeper understanding of what lies behind the various figures and tables and be more confident as what the various reports actually represent. Years of teaching experience suggest that the process of creating a report from scratch—if only once—deepens the user's understanding of what the report actually represents, and can lead them to go beyond the stock

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<sup>1</sup>It is more efficient to perform these analyses using other software, with R or Python being obvious options. However, the advantage of performing the analyses at least once in Excel is that it makes the process completely transparent to the analyst.

reports, requesting custom reports that give even greater insight into buyer behavior.

- Any businessperson working in a firm that does not have anyone who can perform the analyses or the resources to hire a consultant, and must therefore do it themselves. If you are working in a small business or a start-up, this is probably you.
- Students learning to work with customer transaction data. Mastering the simple descriptive analyses associated with a customer-base audit is a good starting point, and such analyses should be undertaken before embarking on more “sophisticated” analyses of buyer behavior.

Note that we will focus on how to create the various figures and tables. We will not spend any time interpreting them, as this is done in TCBA.

## 1.1 Dataset Formats

The dataset used in *TCBA* contains the transaction records for over seven million customers. In order to keep things manageable, we will work with a 1% sample of this dataset — 70,041 customers to be exact.

The raw transaction data for each customer have been aggregated to the quarterly level — for each quarter, we know the number of transactions made by each customer, the associated spend, and the associated profit.<sup>2</sup>

Such a summary will typically be provided to you in either a “wide” or “long” format. With the wide format, we have a separate file for the number of transactions, spend, and profit. Each file has one row per customer and a column for each time period (in this case, quarter). Figure 1.1 illustrates what this looks like for the transaction dataset used in this companion.

The primary downside of this format use that most of the cells are zeroes or empty, and we have to deal with separate files for the number of transactions, spend, and profit. The upside is that we handle one

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<sup>2</sup>If you are performing a customer-base audit on another dataset and have access to the raw transaction data, we recommend that you first create this more aggregated dataset.

|       | A          | B         | C       | D       | E       | F       | O       | P       | Q       | R       |
|-------|------------|-----------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1     | CustomerID | Cohort    | Y2016Q1 | Y2016Q2 | Y2016Q3 | Y2016Q4 | Y2019Q1 | Y2019Q2 | Y2019Q3 | Y2019Q4 |
| 2     | 1          | pre_y2016 | 0       | 1       | 2       | 0       | 0       | 0       | 0       | 0       |
| 3     | 2          | pre_y2016 | 0       | 1       | 1       | 0       | 0       | 0       | 0       | 0       |
| 4     | 3          | pre_y2016 | 0       | 0       | 0       | 0       | 0       | 0       | 0       | 0       |
| 5     | 4          | pre_y2016 | 0       | 0       | 0       | 0       | 0       | 0       | 1       | 0       |
| 6     |            | y2016     | 1       | 0       | 0       | 0       | 0       | 0       | 0       | 0       |
| 70038 | 70038      | y2019_q4  |         |         |         |         |         |         |         | 1       |
| 70040 | 70039      | y2019_q4  |         |         |         |         |         |         |         | 1       |
| 70041 | 70040      | y2019_q4  |         |         |         |         |         |         |         | 2       |
| 70042 | 70041      | y2019_q4  |         |         |         |         |         |         |         | 1       |

**Figure 1.1:** Structure of the wide dataset for transactions

million customers in Excel (assuming one’s computer does not choke on such a large Excel workbook).

With a long dataset, we no longer have one row per customer; we have one row each period (e.g., quarter) each customer was active. If a customer was active in five quarters, there are five rows associated with them. We do not have separate files for the number of transactions, spend, and profit. We simply have a column for each quantity. Figure 1.2 illustrates what this looks like for the dataset used in this companion.

|        | A          | B         | C           | D        | E      | F      |
|--------|------------|-----------|-------------|----------|--------|--------|
| 1      | CustomerID | Cohort    | YearQuarter | NumTrans | Spend  | Profit |
| 2      | 1          | pre_y2016 | y2016_q2    | 1        | 8.16   | 1.5    |
| 3      | 1          | pre_y2016 | y2016_q3    | 2        | 93.48  | 26.32  |
| 4      | 1          | pre_y2016 | y2017_q4    | 2        | 91.84  | 39.66  |
| 5      | 1          | pre_y2016 | y2018_q1    | 2        | 100.64 | 42.21  |
| 6      | 1          | pre_y2016 | y2018_q2    | 2        | 94.25  | 37.15  |
| 7      | 1          | pre_y2016 | y2018_q3    | 2        | 25.13  | 18.95  |
| 130784 | 70036      | y2019_q4  | y2019_q4    | 1        | 28.8   | 10.31  |
| 130785 | 70036      | y2019_q4  | y2019_q4    | 1        | 258.48 | 141.39 |
| 130786 | 70037      | y2019_q4  | y2019_q4    | 1        | 357.84 | 186.57 |
| 130787 | 70038      | y2019_q4  | y2019_q4    | 1        | 131.4  | 68.18  |
| 130788 | 70039      | y2019_q4  | y2019_q4    | 1        | 90     | 41.62  |
| 130789 | 70040      | y2019_q4  | y2019_q4    | 2        | 413.49 | 216.88 |
| 130790 | 70041      | y2019_q4  | y2019_q4    | 1        | 197.4  | 87.35  |

**Figure 1.2:** Structure of the long dataset

We will work with the long dataset, which is provided as a comma-separated values (CSV) file (`cust_data_long.csv`).<sup>3</sup> We also pro-

<sup>3</sup>If this file format is not automatically opened by Excel on your computer, it is very easy to import it into Excel.

vide the same 1% sample in a wide format (`cust_by_qtr_trans.csv`, `cust_by_qtr_spend.csv`, and `cust_by_qtr_profit.csv`) and describe in Appendix A how to create the various working datasets.

## 1.2 How to Use This Companion

First and foremost, we assume you have read *TCBA*. If you have not, stop right now, read the book, and only then come back to this document.

It is recommended that you work your way through this companion starting with a blank Excel workbook. The “solution” spreadsheets should be viewed as model answers, against which you can compare your analyses. (Simply looking through them is no substitute for creating all the reports for yourself.)

It is assumed that you have some familiarity with pivot tables in Excel. (If not, help can be found on the Microsoft Office support website or via various tutorials on YouTube.) Some readers will be able to look at a completed report and quickly work out how it was created. For the rest of you, this note provides detailed step-by-step instructions. If you are an Excel novice, the process of creating all the reports for yourself should help develop your Excel skills, independent of the application context considered here. Note that we are not presenting *the* definitive way of creating any given report. In many cases, there are several different ways of conducting the same analysis, some of which will be “better” than the approach presented here.

It is also assumed that you know how to “tidy up” the charts generated by Excel (e.g., change the format of the axes). The Excel-generated figures presented in this documents have received such treatment.

Finally, there are frequently multiple ways of performing certain analyses in Excel. The approaches taken here should not be viewed as definitive. Furthermore, Excel is constantly evolving, with new functions being added each year. Some of the functions used in this note will become outdated and there will be “better” ways to perform the analysis. The challenge for the reader is how to learn about the new features being added to Excel.

## Typographical Conventions

The following typographical conventions are used in this note:

### **Bold**

Used for the names of Excel worksheets.

### Underlined

Used for the name of a column heading in a worksheet.

### `Constant width`

Used for formulas and file names.

# Chapter 2

## Lens 1 Analyses

The primary objective of Lens 1 analyses is to understand how different our customers are from one another. In this chapter we show how to perform all the analyses presented in Chapter 3 of *TCBA*.

### 2.1 Creating the Working Dataset

All the figures and tables that we use to summarize how customers differ in their buying behavior in a given period of time—in our case, a calendar year—can be created from a dataset that reports, for each customer, the number of transactions they made, their total spend, and (assuming cost data are available) the profit associated with their purchases in that period.

We wish to summarize the variability in buying behavior of Madrigal’s customers in 2019. As our master dataset summarizes each customer’s buying at the quarterly level, we first need to create a working dataset that aggregates the quarterly numbers to the annual level for each customer that made at least one transaction with Madrigal in 2019. An extract of the required working dataset is given in Figure 2.1.

We can create this dataset in the following manner.

- We open `cust_data_long.csv` in Excel and save it as an `xlsx` file (e.g., `creating_lens_1_data.xlsx`).
- As we will be aggregating the data up to the annual level, we need to identify the year associated with each record. We have a `YearQuarter` variable in column C. To create a `Year` variable, we first enter “Year” in cell G1. Next we enter `=LEFT(C2,5)` in cell

|       | A          | B        | C       | D      |
|-------|------------|----------|---------|--------|
| 1     | CustomerID | NumTrans | Spend   | Profit |
| 2     | 4          | 1        | 58.78   | 30.47  |
| 3     | 8          | 6        | 239.96  | 134.75 |
| 4     | 9          | 3        | 74.58   | 33.91  |
| 5     | 10         | 13       | 1778.24 | 911.73 |
| 6     | 11         | 1        | 109.8   | 48.63  |
| 7     | 12         | 6        | 478.3   | 223.47 |
| 8     | 22         | 6        | 162.46  | 48.98  |
| 9     | 24         | 1        | 48      | 14.08  |
| 10    | 28         | 2        | 234.72  | 102.2  |
| 11    | 30         | 3        | 119.02  | 52.15  |
| 31853 | 70038      | 1        | 131.4   | 68.18  |
| 31854 | 70039      | 1        | 90      | 41.62  |
| 31855 | 70040      | 2        | 413.49  | 216.88 |
| 31856 | 70041      | 1        | 197.4   | 87.35  |

**Figure 2.1:** Working dataset for Lens 1 analyses

G2 and copy this formula down to the last row of the dataset (cell G130790).

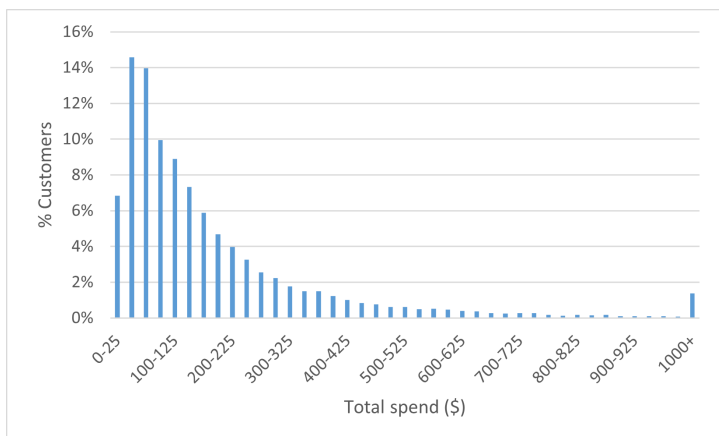
- In order to create the annual summaries, we insert a pivot table where Rows is CustomerID and Values is (Sum of) NumTrans, Spend, and Profit. As we only wish to compute this summary for 2019, we drag Year to Filters and select y2019. This creates the **Sheet 1** worksheet.
- In order to keep our Excel workbook a manageable size, we copy the contents of the pivot table (A4:D31858) into a new workbook, leaving the first row empty. We add the following variable names in cells A1:D1: CustomerID, NumTrans, Spend, and Profit. This gives us the dataset reported in Figure 2.1.
- We rename the worksheet **Data**. (Don't forget to save this new workbook, say `lens_1_analyses.xlsx`)

We see that 31,855 individuals made at least one purchase from Madrigal in 2019. Summing the NumTrans column, we see that these individuals made a total of 60,730 transactions. The sum of the Spend column is \$5.8 million and the sum of the Profit column is \$2.8 million. This implies that the average number of transactions per active customer is 1.9, and that the average spend and average profit numbers are \$183 and \$88, respectively. But how much variability is there around these average numbers?

## 2.2 Foundational Plots

### 2.2.1 The Distribution of Spend

Our task is to create Figure 2.2, which reports the distribution of spend across the 31,855 individuals that made at least one purchase from Madrigal in 2019.



**Figure 2.2:** Distribution of total spend in 2019

The first step in creating such a figure is to determine the binning strategy. This requires us to specify the bin width and the number of bins. The distributions of most customer behaviors are skewed with a long right tail, typically a very long tail, with the maximum 10 or 100 times the mean, if not larger. It therefore usually makes sense to have a “greater than  $x$ ” bin, \$1000 in Figure 2.2. This is sometimes called the right-censoring point.

These choices depend on the data being plotted, the key message(s) we wish to convey via the figure, and the “look” of the figure. For example, a very narrow bin width will result in a “noisy” plot (e.g., a bin width of \$2 in the above example), while a very wide bin width will hide important information about the nature of the variability in the quantity being plotted (e.g., a bin width of \$100 in the above example). While any number could be used as the width of the bin, we would tend to choose 1, 2, 5, 10, 20, 25, 50, 100, 200, 250, 500, ....

We make the binning decision by looking at some descriptive statistics.

- We copy the CustomerID and Spend columns into a new worksheet, which we rename **Spend distribution**.
- We compute some basic descriptive statistics.
  - The minimum total spend is computed in cell E2:  
=MIN(B2:B31856)
  - The maximum total spend is computed in cell E3:  
=MAX(B2:B31856)
  - The average total spend is computed in cell E4:  
=AVERAGE(B2:B31856)
  - The median total spend is computed in cell E5:  
=MEDIAN(B2:B31856)
- We see that the maximum spend in 2019 (\$6695) is almost 37 times the mean spend (\$183). We also note that the mean is larger than the median (\$113). This begs the question, “What percentage of customers have below average spend?” The answer is 69%, as computed by entering  
=COUNTIF(B2:B31856, "<"&E4)/COUNT(B2:B31856)  
in cell E6.
- There appears to be quite a bit of variability in 2019 spend. To get further insight into the distribution of spend across customers, we use Excel’s percentile function to determine the spend quantities associated with the various percentile levels.
  - We enter 0.05 in cell D8, =D8+0.05 in cell D9, and copy this formula down to D26. Cells D8:D26 are then formatted as percentages.
  - The PERCENTILE.INC function returns the k-th percentile of values in a range. We enter  
=PERCENTILE.INC(B\$2:B\$31856,D8)  
in cell E8 and copy this formula down to E26.
- We see that 5% of customers spent \$22.20 or less during the year, 10% spent \$30.22 or less, and so on. The heaviest 5% each spent more than \$578.52 in 2019.

This would suggest that a bin width of \$20 or \$25 is probably appropriate. We want the plot to convey the skewed nature of the

distribution and therefore the percentage of the observations in the final bin should not be too high. Noting that 5% of customers spent more than \$578, choosing \$600 as the cut-off point would probably result in too many observations in the final bin. We will try \$1000. We will go with a bin width of \$25 as this results in 41 bins. (A bin width of \$20 would result in 51 bins.)

Having decided on the bin width and number of bins, the next step is to determine how many customers fall into each bin.

- Staying in **Spend Distribution**, we first create the bin boundaries. We enter 25 in cell G2, =G2+25 in cell G3, and copy this formula down to cell G41. The upper limit is entered in cell G42: =MAX(B2:B31856).
- For each boundary point, we want to compute the number of customers whose spend was at or below that number. We enter =COUNTIF(B\$2:B\$31856,"<="&G2) in cell H2 and copy the formula down to cell H42.
- In order to compute the number of customers falling into each bin, we first enter =H2 in cell I2. Next we enter =H3-H2 in cell I3 and copy the formula down to cell I42.
- We convert these counts into relative frequencies by entering =I2/SUM(I\$2:I\$42) in cell J2 and copying the formula down to cell J42. Cells J2:J42 are then formatted as percentages.
- Plotting these relative frequencies gives us Figure 2.2.

Reflecting on the calculations behind this plot, the \$25–50 bin counts those customers whose spend was greater than \$25 and less than or equal to \$50. (The formal notation for this is  $(25, 50]$ .) Now, the dataset contains two customers with a spend of \$0 in 2019. (How do we know this? =COUNTIF(B2:B31856,"=0").) The binning approach used above means that we have included these two customers in the first bin.

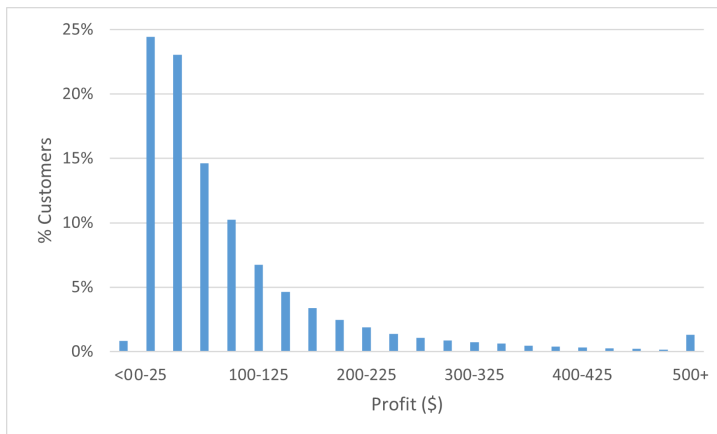
If you are familiar with Excel, you might be asking why we did not compute the number of customers falling into each bin using the **FREQUENCY** function (or equivalently the histogram feature in the Data Analysis ToolPak. We choose not to as (at the time of writing) there is a bug associated with the function that sometimes results in incorrect frequency counts.

### Exercise

Try out other bins widths and cut-off points. In your judgment, does this result in a more visually appealing plot that conveys the point that there is a lot of variability on customer spend and that the notion of an “average customer” makes no sense?

## 2.2.2 The Distribution of Customer Profit

Our task is to create Figure 2.3, which reports the distribution of profit across these same 31,855 customers. The basic structure of the plot is the same as that for the distribution of spend with one key difference: we have a <\$0 bin to account for the loss-making customers.



**Figure 2.3:** Distribution of customer profit in 2019

The steps associated with the creation of this plot are basically the same as those used to create Figure 2.2. First, we need to choose our binning strategy and so we start by looking at some descriptive statistics.

- We copy the CustomerID and Profit columns from **Data** into a new worksheet, which we rename **Profit distribution**.
- We compute some basic descriptive statistics.
  - The minimum total spend is computed in cell E2:  
`=MIN(B2:B31856)`

- The maximum total spend is computed in cell E3:  
=MAX(B2:B31856)
- The average total spend is computed in cell E4:  
=AVERAGE(B2:B31856)
- The median total spend is computed in cell E5:  
=MEDIAN(B2:B31856)
- We see that customer-level profit in 2019 ranged from –\$652 to \$3347. The mean (\$88) is larger than the median (\$54), with 69% of the customers having below average profit:  
=COUNTIF(B2:B31856,"<"&E4)/COUNT(B2:B31856)
- To get further insight into the distribution of profit across customers, we determine the profit numbers associated with the various percentile levels.
  - We enter 0.05 in cell D8, =D8+0.05 in cell D9, and copy this formula down to D26. Cells D8:D26 are then formatted as percentages.
  - We enter =PERCENTILE.INC(B\$2:B\$31856,D8) in cell E8 and copy this formula down to E26.
- We see that 5% of customers had a 2019 profit at or below \$7.70, 10% had a profit at or below \$12.47, and so on. A 2019 profit greater than \$282.45 put the customer in the top 5% of customers.

The mean, median and maximum profits are around 45–50% of the corresponding spend numbers. We will therefore try \$500 as the cut-off point for the final bin. We will stay with a \$25 bin width, although a good case could be made for \$20, if not \$10.

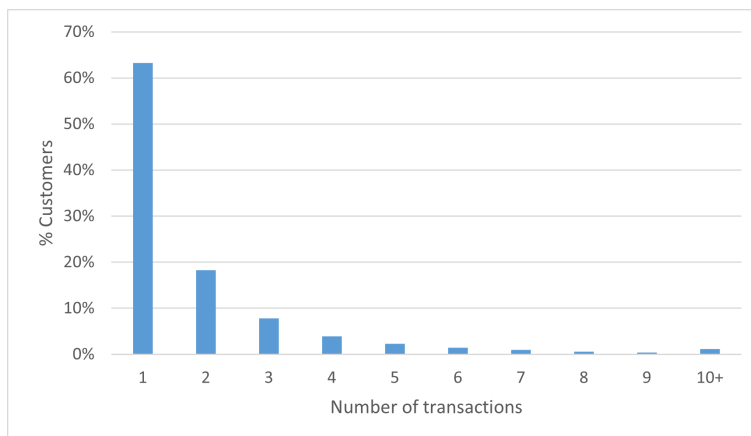
The next step is to determine how many customers fall into each bin. Compared to the steps associated with creating the spend distribution, the key difference is the addition of a bin for the loss-making customers.

- Staying in **Profit Distribution**, we first create the bin boundaries. We enter 0 in cell G2, =G2+25 in cell G3, and copy this formula down to cell G22. The upper limit is entered in cell G23: =MAX(B2:B31856).

- For each boundary point, we want to compute the number of customers whose profit was at or below that number. We enter `=COUNTIF(B$2:B$31856,"<="&G2)` in cell H2 and copy the formula down to cell H23.
- In order to compute the number of customers falling into each bin, we first enter `=H2` in cell I2. Next we enter `=H3-H2` in cell I3 and copy the formula down to cell I23.
- We convert these counts into relative frequencies by entering `=I2/SUM(I$2:I$23)` in cell J2 and copying the formula down to cell J23. Cells J2:J23 are then formatted as percentages.
- Plotting these relative frequencies gives us Figure 2.3.

### 2.2.3 The Distribution of the Number of Transactions

Our task is to create Figure 2.4, which reports the distribution of the number of transactions made by the 31,855 individuals that made at least one purchase from Madrigal in 2019.



**Figure 2.4:** Distribution of the number of transactions in 2019

We can create this plot in the following manner.

- The easiest way to compute the number of customers that made 1, 2, 3, ... transactions is to use Excel's PivotTable tool. Going back to **Data** and with the active cell somewhere in the main table, we insert a pivot table where Rows is NumTrans and

Values is (Count of) CustomerID. We rename the resulting worksheet **Transactions distribution**.

- We see that the distribution of the number of transactions is again right skewed, with one customer making 58 transactions. Recall from above that the average number of transactions made by a 2019 customer is 1.9. With 63% of the customers making just one purchase (20,149/31,855), the mean is clearly a misleading summary measure.
- The next step is to choose a right-censoring point for the plot. Assuming we plan to plot the frequencies associated with 1, 2, 3, ... transactions, we would probably choose a number that is a multiple of five. In this case, five seems to be too low, with 6.8% of the customers making five or more transactions in 2019. We therefore go with ten.
- We copy cells A4:B12 to D4:E12. Next, we enter “10+” in cell D13 and =B43-SUM(E4:E12) in cell E13 to determine the number of customers that made ten or more transactions.
- Finally, we convert these counts into relative frequencies by entering =E4/SUM(E\$4:E\$13) in cell F2 and copying the formula down to cell F13. Cells F2:F13 are then formatted as percentages.
- Plotting these relative frequencies gives us Figure 2.4.

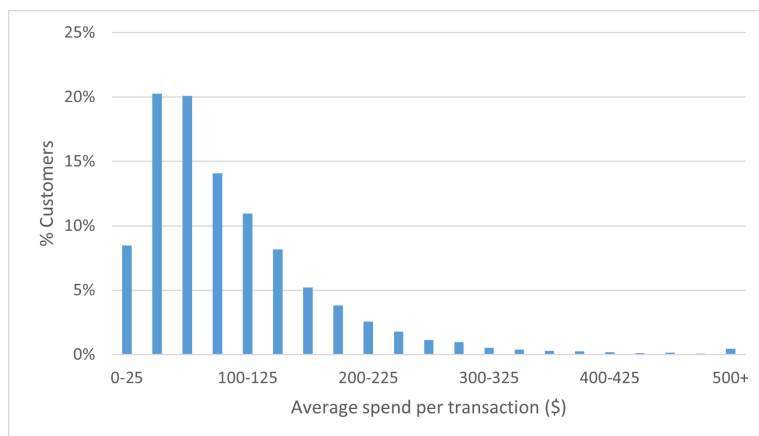
In most cases, the distribution of the number of transactions is reverse-J-shaped (i.e., the most frequent count is 1, and followed by 2, with subsequent counts getting smaller and smaller). Occasionally, we will observe a mode away from 1. In the overwhelming majority of cases, we will plot the number of transactions using a bin width of 1. In some rare circumstances, where the mean and median number of transactions are high, we may choose to use a different bin width. However, it is important that, with the exception of the right-most bin, the bins are of equal width. For example, we would discourage the creation of a plot of the number of customers making 1 transaction, 2–4 transactions, 5–9 transactions, and 10+ transactions. If, for some reason, you wish to communicate the data in such a manner, a table would be preferable.

## 2.3 Further Analyses

Additional insights into how customers differ in their buying behavior can be obtained by examining quantities that can be derived from the spend, profit, and transaction numbers. We will focus on two: average spend (per transaction) and (average) margin. Each customer's average spend (per transaction) is simply their total spend in the given time period (in this case, the whole of 2019) divided by the number of transactions they made in that period. Similarly, (average) margin is their profit for that period divided by their spend.

### 2.3.1 The Distribution of Average Spend per Transaction

Our task is to create Figure 2.5, which shows the distribution of average spend using bins that are \$25 wide.



**Figure 2.5:** Distribution of average spend per transaction in 2019

We follow what should now be a familiar process.

- The first step is to compute each customer's average spend. We make a copy of **Data**, renaming it **Average spend**, and delete the Profit column.
- Having entered the column heading "Avg spend" in cell D1, we enter `=C2/B2` in cell D2 and copy this formula down to cell D31856.

- We compute some basic descriptive statistics.
  - The minimum total spend is computed in cell G2:  
=MIN(D2:D31856)
  - The maximum total spend is computed in cell G3:  
=MAX(D2:D31856)
  - The average total spend is computed in cell G4:  
=AVERAGE(D2:D31856)
  - The median total spend is computed in cell G5:  
=MEDIAN(D2:D31856)
  - The percentage of customers whose average spend is below average is computed in cell G6:  
=COUNTIF(D2:D31856,"<"&G4)/COUNT(D2:D31856)
- To get further insight into the distribution of spend across customers, we determine the average spend numbers associated with the various percentile levels.
  - We enter 0.05 in cell F8, =F8+0.05 in cell F9, and copy this formula down to cell F26. Cells F8:F26 are then formatted as percentages.
  - We enter =PERCENTILE.INC(D\$2:D\$31856,F8) in cell G8 and copy this formula down to cell G26.

We see evidence of the now familiar right-skewed distribution. We will use a bin width of \$25 and choose \$500 as the cut-off point for the final bin. However, a good case could be made for, say, a \$20 bin width and a cut-off of \$300. One advantage of a \$500 cut-off is that it does provide a better sense of the long tail of this distribution.

The next step is to determine how many customers fall into each average spend bin.

- We create the bin boundaries by entering 25 in cell I2, =G2+25 in cell I3, and copying this formula down to cell I21. The upper limit is entered in cell I22: =MAX(D2:D31856).
- For each boundary point, we want to compute the number of customers whose spend was at or below that number. We enter =COUNTIF(D\$2:D\$31856,"<="&I2) in cell J2 and copy the formula down to cell J22.

- In order to compute the number of customers falling into each bin, we first enter =J2 in cell K2. Next we enter =J3-J2 in cell K3 and copy the formula down to cell K22.
- We convert these counts into relative frequencies by entering =K2/SUM(K\$2:K\$22) in cell L2 and copying the formula down to cell L22. Cells L2:L22 are then formatted as percentages.
- Plotting these relative frequencies gives us Figure 2.5.

### A statistical aside

Recall that the average spend per 2019 customer is \$183 and the average number of transactions is 1.9. We could be tempted to say that it follows that the average spend per transaction is \$183/1.9 = \$96. However, we have just seen that mean average spend per transaction is \$99. How do we explain the difference in these two numbers?

Let  $\text{spend}_i$  and  $\text{trans}_i$  denote customer  $i$ 's spend and number of transactions in a given period (2019 in this specific example). Assuming we have  $I$  customers, the formula behind the \$96 number is

$$\begin{aligned} \frac{\sum_{i=1}^I \text{spend}_i}{I} \bigg/ \frac{\sum_{i=1}^I \text{trans}_i}{I} &= \frac{\sum_{i=1}^I \text{spend}_i}{\sum_{i=1}^I \text{trans}_i} \\ &= \frac{\sum_{i=1}^I \text{spend}_i \times \frac{\text{trans}_i}{\text{trans}_i}}{\sum_{i=1}^I \text{trans}_i} \\ &= \sum_{i=1}^I \left( \frac{\text{trans}_i}{\sum_{i=1}^I \text{trans}_i} \right) \frac{\text{spend}_i}{\text{trans}_i}, \end{aligned}$$

while the formula behind the \$99 number is

$$\frac{1}{I} \sum_{i=1}^I \frac{\text{spend}_i}{\text{trans}_i}.$$

In other words, \$99 is the unweighted-average average spend per transaction and \$96 is the weighted-average average spend per transaction where the weight depends of the number of transac-

tions made by each customer. Clearly these two quantities are equal only when  $\text{trans}_i$  is the same for all  $I$  customers.

Whenever you see average average spend (per transaction) numbers in any report, it is important that you understand how they are being calculated. In what follows, we will call the first quantity Average Order Value (AOV) to avoid this confusion.

### 2.3.2 Further Analysis of Average Spend per Transaction

Given the variability in both the number of transactions and average spend per transaction across customers, it is natural to ask whether these two quantities are related. In *TCBA*, we reported the mean and median average spend per transaction by transaction level. We present an extended version of this in Table 2.1, which is created in the following manner.

|    | G       | H     | I       | J    | K       | L    | M      | N     |
|----|---------|-------|---------|------|---------|------|--------|-------|
| 1  | # trans | Mean  | Std dev | Min  | Max     | 5%   | Median | 95%   |
| 2  | 1       | \$100 | \$91    | \$0  | \$3,451 | \$18 | \$72   | \$252 |
| 3  | 2       | \$100 | \$83    | \$7  | \$2,545 | \$24 | \$82   | \$232 |
| 4  | 3       | \$99  | \$69    | \$8  | \$810   | \$27 | \$84   | \$218 |
| 5  | 4       | \$97  | \$66    | \$9  | \$724   | \$28 | \$83   | \$206 |
| 6  | 5       | \$91  | \$54    | \$10 | \$400   | \$27 | \$78   | \$191 |
| 7  | 6       | \$91  | \$55    | \$11 | \$415   | \$26 | \$81   | \$193 |
| 8  | 7       | \$87  | \$49    | \$11 | \$278   | \$27 | \$77   | \$180 |
| 9  | 8       | \$88  | \$50    | \$11 | \$340   | \$29 | \$78   | \$186 |
| 10 | 9       | \$82  | \$44    | \$11 | \$276   | \$24 | \$72   | \$160 |
| 11 | 10+     | \$83  | \$44    | \$12 | \$268   | \$29 | \$74   | \$168 |

**Table 2.1:** Analysis of average spend per transaction by transaction level

- We make a copy of **Average spend**, rename it **Average spend by num trans**, and delete columns F:L.
- We will use the same frequency bins as for Figure 2.4. We create these by entering “Trans bin” in cell E1, `=IF(B2>=10, "10+", B2)` in cell E2, and copying this formula down to cell E31856.
- We enter the bin labels (1, 2, ..., 9, 10+) in cells G2:G11.
- We wish to compute the mean average spend for those that made just one transaction in 2019. We can do this by combin-

ing the **AVERAGE** and **IF** functions,<sup>1</sup> entering **=AVERAGE(IF(E\$2:E\$31856=G2,D\$2:D\$31856))** in cell H2. Copying this formula down to cell H11 computes the mean average spend for the other transaction levels.<sup>2</sup>

- We compute the other quantities of interest in a similar manner for those customers that made one transaction in 2019:
  - The standard deviation of average spend is computed in cell I2:  
`=STDEV.S(IF(E$2:E$31856=G2,D$2:D$31856))`
  - The minimum average spend is computed in cell J2:  
`=MIN(IF(E$2:E$31856=G2,D$2:D$31856))`
  - The maximum average spend is computed in cell K2:  
`=MAX(IF(E$2:E$31856=G2,D$2:D$31856))`
  - The average spend associated with the 5th percentile is computed in cell L2:  
`=PERCENTILE.INC(IF(E$2:E$31856=G2,D$2:D$31856),0.05)`
  - The median average spend is computed in cell M2:  
`=MEDIAN(IF(E$2:E$31856=G2,D$2:D$31856))`
  - The average spend associated with the 95th percentile is computed in cell N2:  
`=PERCENTILE.INC(IF(E$2:E$31856 =G2,D$2:D$31856),0.95)`

Copying cells G2:N2 down to row 11 computes these quantities for the other transaction levels.

Note that if we had just been interested in the first four quantities, we could have created this table using a pivot table, with Rows being Trans bin and Values being Average, StdDev, Min, and Max of Avg spend. However, in order to compute the percentiles, we had to combine the relevant function with the **IF** function.

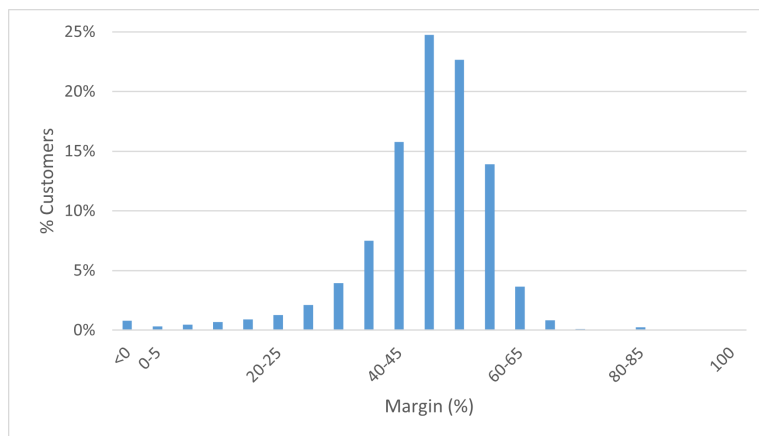
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<sup>1</sup>We could also use the **AVERAGEIF** function.

<sup>2</sup>If you are using a recent version of Excel for Microsoft 365 or (at least) Excel 2021), this will work automatically. If you are using an older version of Excel, it is time to upgrade. Failing that, as this is an array formula, you will need to press Ctrl+Shift+Enter. This has to be done for each cell.

### 2.3.3 The Distribution of Average Margin

Our task is to create Figure 2.6, which reports the distribution of average customer profit margin across the 31,855 individuals that made at least one purchase from Madrigal in 2019.



**Figure 2.6:** Distribution of average margin in 2019

We create this plot in the following manner.

- The first step is to compute each customer’s (average) margin. We add a new worksheet, renaming it **Margin** and copy the contents of the CustomerID column from **Data** into column A.
- We need to remember that we have two customers with zero spend. So as to avoid the **#DIV/0!** error, which will cause problems in the subsequent analysis, we compute a customer’s average margin in the following manner. Having entered the column heading “Margin” in cell B1, we enter `=IF(Data!C2>0,Data!D2/Data!C2,"")` in cell B2 and copy this formula down to cell B31856.<sup>3</sup>

Note that this average margin number we have just computed is not the average of the margins associated with each of the customer’s individual transactions; rather, it is the overall margin

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<sup>3</sup>A note to the novice Excel user: When entering this formula, we do not actually type `=IF(Data!C2>0,Data!D2/Data!C2,"")`. Having typed `=IF` (in cell B2 (but not pressing `<Enter>`), we then go to the **Data** worksheet, click on cell C2 and type `>0,`. Then we click on cell D2, type `/`, click on cell C2, and type `,"")` before pressing `<Enter>`.

across all their purchasing in 2019. As our “raw” dataset is aggregated to the quarterly level, we cannot compute transaction-level margin numbers for any customer who made more than one purchase from Madrigal in any quarter.

- We compute the basic descriptive statistics and the margins associated with different percentiles.
  - The minimum margin is computed in cell E2:  
`=MIN(B2:B31856)`
  - The maximum margin is computed in cell E3:  
`=MAX(B2:B31856)`
  - The average margin is computed in cell E4:  
`=AVERAGE(B2:B31856)`
  - The median margin is computed in cell E5:  
`=MEDIAN(B2:B31856)`
  - The percentage of customers whose margin is below average is computed in cell E6:  
`=COUNTIF(B2:B31856,"<"&E4)/COUNT(B2:B31856)`
- To get further insight into the distribution of margin across customers, we determine the margin numbers associated with the various percentile levels.
  - We enter 0.05 in cell D8, `=D8+0.05` in cell D9, and copy this formula down to cell D26. Cells D8:D26 are then formatted as percentages.
  - We enter `=PERCENTILE.INC(B$2:B$31856,D8)` in cell E8 and copy this formula down to cell E26.

We note that, in contrast to the other four distributions we have considered, the distribution of margin is left-skewed.

Given the range of margins we need to convey with the figure, we will use margin bin widths of 5%, with a <0% bin to account for the loss-making customers.

- We create the bin boundaries by entering 0 in cell G2, `=G2+0.05` in cell G3, and copying this formula down to cell G22.
- For each boundary point, we want to compute the number of customers whose margin was at or below that number. We

enter `=COUNTIF(B$2:B$31856,"<="&G2)` in cell H2 and copy the formula down to cell H22.

- In order to compute the number of customers falling into each bin, we first enter `=H2` in cell I2. Next we enter `=H3-H2` in cell I3 and copy the formula down to cell I22.
- We convert these counts into relative frequencies by entering `=I2/SUM(I$2:I$22)` in cell J2 and copying the formula down to cell J22. Cells J2:J22 are then formatted as percentages.

Plotting these relative frequencies gives us Figure 2.6. We have chosen to include the (85%, 90%], (90%, 95%], and (95%, 100%] bins, even though they contain zero observations. You could choose not to include them.

## 2.4 Decile Analyses

We have looked at three fundamental behavioral measures (spend, profit, and transactions) and two derived measures (average spend and margin). The key conclusion is that customers differ in their behavior and therefore their value to the firm. With the exception of margin, we see that the majority of customers are below average. This means a small group of the firm's customers account for a sizable chunk of its revenues and profits.

A useful way of communicating this is via a decile analysis which, as the name suggests, sees us dividing the customer base into 10 equally sized groups and then summarizing behavior by group. There are two standard approaches to group formation: having sorted customers by their profitability in the period of interest, the first approach creates groups that account for 10% of the customers. The first decile represents the 10% of customers that are most profitable, and so on. This generates what we will call the customer decile report. The second approach creates groups of customers such that each group accounts for 10% of the total customer profit, resulting in what we will call the profit decile report.

### 2.4.1 The Customer Decile Report

Our task is to create Table 2.2, which gives us insight into the underlying behaviour of the customer-based deciles (i.e., each decile represents

10% of the customers active in 2019).<sup>4</sup>

|    | G      | H      | I       | J       | K        | L              | M               | N   | O        | P          |
|----|--------|--------|---------|---------|----------|----------------|-----------------|-----|----------|------------|
| 3  | Decile | % Cust | % Trans | % Spend | % Profit | Avg spend/cust | Avg profit/cust | AOF | AOV      | Avg margin |
| 4  | 1      | 10%    | 27%     | 38%     | 40%      | \$700.50       | \$348.93        | 5.1 | \$138.14 | 50%        |
| 5  | 2      | 10%    | 14%     | 17%     | 18%      | \$315.03       | \$155.44        | 2.7 | \$118.61 | 49%        |
| 6  | 3      | 10%    | 11%     | 12%     | 12%      | \$217.75       | \$106.70        | 2.0 | \$107.05 | 49%        |
| 7  | 4      | 10%    | 9%      | 9%      | 9%       | \$165.25       | \$79.72         | 1.7 | \$96.96  | 48%        |
| 8  | 5      | 10%    | 8%      | 7%      | 7%       | \$129.31       | \$60.55         | 1.5 | \$83.64  | 47%        |
| 9  | 6      | 10%    | 7%      | 5%      | 5%       | \$99.14        | \$45.69         | 1.4 | \$72.42  | 46%        |
| 10 | 7      | 10%    | 7%      | 4%      | 4%       | \$75.89        | \$34.05         | 1.3 | \$60.00  | 45%        |
| 11 | 8      | 10%    | 6%      | 3%      | 3%       | \$57.41        | \$24.88         | 1.2 | \$48.51  | 43%        |
| 12 | 9      | 10%    | 6%      | 2%      | 2%       | \$41.27        | \$16.66         | 1.1 | \$36.94  | 40%        |
| 13 | 10     | 10%    | 6%      | 2%      | 1%       | \$30.64        | \$5.98          | 1.1 | \$27.41  | 20%        |
| 14 |        |        |         |         |          | \$183.23       | \$87.86         | 1.9 | \$96.11  | 48%        |

**Table 2.2:** Customer decile report

The first step is to create the summary table given in Table 2.3, which reports, for each decile, the number of customers in each decile and the total number of transactions, total spend, and total profit associated with each decile.

|    | A                  | B                   | C               | D                 | E                 |
|----|--------------------|---------------------|-----------------|-------------------|-------------------|
| 3  | Row Labels         | Count of CustomerID | Sum of NumTrans | Sum of Spend      | Sum of Profit     |
| 4  | 1                  | 3186                | 16156           | 2231779.57        | 1111697.17        |
| 5  | 2                  | 3185                | 8459            | 1003364.24        | 495070.84         |
| 6  | 3                  | 3186                | 6481            | 693761.37         | 339932.77         |
| 7  | 4                  | 3185                | 5428            | 526321.06         | 253914.4          |
| 8  | 5                  | 3186                | 4926            | 411989.41         | 192920.74         |
| 9  | 6                  | 3185                | 4360            | 315766.81         | 145536.06         |
| 10 | 7                  | 3186                | 4030            | 241787.37         | 108468.06         |
| 11 | 8                  | 3185                | 3770            | 182866.58         | 79247.46          |
| 12 | 9                  | 3186                | 3560            | 131499.96         | 53084.36          |
| 13 | 10                 | 3185                | 3560            | 97575.99          | 19031.82          |
| 14 | <b>Grand Total</b> | <b>31855</b>        | <b>60730</b>    | <b>5836712.36</b> | <b>2798903.68</b> |

**Table 2.3:** Summary of 2019 customer behavior by customer decile

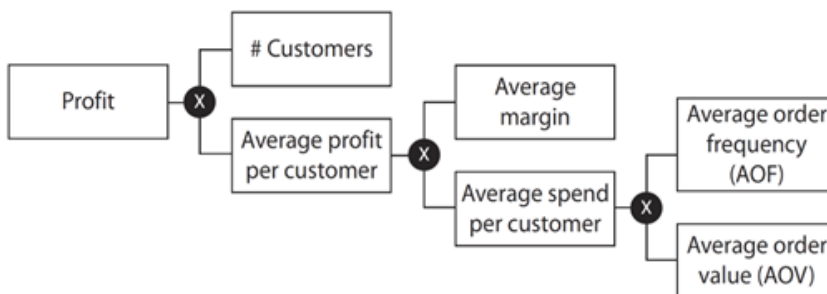
This summary table is created in the following manner.

- We make a copy of **Data**, renaming it **Decile analysis (i)**.
- We need to assign a decile label to each customer, where 1 is assigned to the 10% most profitable customers . . . and 10 is assigned to the 10% least profitable customers.

<sup>4</sup>This table is a combined version of *TCBA* Tables 3.2 and 3.3 without the raw numbers (which are given in Table 2.3 below).

- We start by sorting the data by Profit from Largest to Smallest.
- As an intermediate step to creating the decile label, we create a rank variable, where the rank of 1 is assigned to the most profitable customer, 2 to the second most profitable customer, and so on. We enter 1 in cell E2, =E2+1 in cell E3, and copy the formula down to cell E31856.
- The next step is to convert the rank numbers into a decile label. We enter “Decile” in cell F1, =INT(10\*(E2-1)/E\$31856)+1 in cell F2, and copy the formula down to cell F31856. (Take time to reflect on the logic of the formula. For example, why do we have E2-1 and not just E2?)
- Table 2.3 is created by inserting a pivot table where Rows is Decile and Values is (Count of) CustomerID, and (Sum of) NumTrans, Spend, and Profit. We rename the resulting worksheet **Decile analysis (Customer)**.

We create Table 2.2 by applying the profit decomposition given in Figure 2.7 to the data summary reported in Table 2.3. (See *TCBA* (p. 49) for the derivation of this decomposition.)



**Figure 2.7:** A multiplicative decomposition of profit

- Staying within the **Decile analysis (Customer)** worksheet, we enter the decile labels 1 to 10 in cells G4:G13.
- The percentage of customers associated with decile 1 is computed by entering =B4/B\$14 in cell H4.

- The percentage of the firm's total number of transactions made by customers assigned to decile 1 is computed by entering `=C4/C$14` in cell I4.
- The percentage of the firm's revenue attributable to customers assigned to decile 1 is computed by entering `=D4/D$14` in cell J4.
- The percentage of the firm's customer profit associated with members of the first decile is computed by entering `=E4/E$14` in cell K4.
- The average spend per decile 1 customer is computed by entering `=D4/B4` in cell L4.
- The average profit per decile 1 customer is computed by entering `=E4/B4` in cell M4.
- The average order frequency is computed by entering `=C4/B4` in cell N4.
- The average order value is computed by entering `=D4/C4` in cell O4.
- The average margin for decile 1 customers is computed by entering `=E4/D4` in cell P4.
- Having applied the appropriate formatting to each cell, we copy the formulas in cells H4:P4 down to row 13.
- We compute the overall values of average spend per customer, average profit per customer, AOF, AOV, and average margin by copying cells L13:P13 to L14:P14.

### Exercise

Suppose we do not have customer-level profit numbers, only the number of transactions and total spend associated with each customer. Perform a decile analysis of Madrigal's revenue in 2019.

## 2.4.2 The Profit Decile Report

The alternative way of performing a decile analysis is to create groups of customers such that each group accounts for 10% of the total customer profit, resulting in Table 2.4.

|    | G      | H      | I       | J       | K        | L              | M               | N    | O              | P   |
|----|--------|--------|---------|---------|----------|----------------|-----------------|------|----------------|-----|
| 3  | Decile | % Cust | % Trans | % Spend | % Profit | Avg spend/cust | Avg profit/cust | AOF  | AOV Avg margin |     |
| 4  | 1      | 1%     | 6%      | 10%     | 10%      | \$1,625.96     | \$813.31        | 10.2 | \$160.13       | 50% |
| 5  | 2      | 2%     | 7%      | 10%     | 10%      | \$843.15       | \$419.97        | 6.0  | \$140.10       | 50% |
| 6  | 3      | 3%     | 7%      | 10%     | 10%      | \$594.67       | \$295.34        | 4.5  | \$130.77       | 50% |
| 7  | 4      | 4%     | 7%      | 10%     | 10%      | \$443.74       | \$220.86        | 3.5  | \$125.62       | 50% |
| 8  | 5      | 5%     | 8%      | 10%     | 10%      | \$347.30       | \$170.92        | 2.9  | \$120.96       | 49% |
| 9  | 6      | 7%     | 8%      | 10%     | 10%      | \$269.53       | \$132.97        | 2.3  | \$114.99       | 49% |
| 10 | 7      | 9%     | 9%      | 10%     | 10%      | \$209.42       | \$102.67        | 2.0  | \$105.31       | 49% |
| 11 | 8      | 11%    | 10%     | 10%     | 10%      | \$161.14       | \$77.43         | 1.7  | \$95.72        | 48% |
| 12 | 9      | 17%    | 13%     | 10%     | 10%      | \$113.67       | \$52.79         | 1.5  | \$77.94        | 46% |
| 13 | 10     | 42%    | 26%     | 12%     | 10%      | \$52.66        | \$21.14         | 1.2  | \$44.81        | 40% |
| 14 |        |        |         |         |          | \$183.23       | \$87.86         | 1.9  | \$96.11        | 48% |

**Table 2.4:** Profit decile report

The first step is to create the summary table given in Table 2.5, which reports, for each decile, the number of customers in each decile and the total number of transactions, total spend, and total profit associated with each decile.

|    | A                  | B                   | C               | D                 | E                 |
|----|--------------------|---------------------|-----------------|-------------------|-------------------|
| 3  | Row Labels         | Count of CustomerID | Sum of NumTrans | Sum of Spend      | Sum of Profit     |
| 4  | 1                  | 344                 | 3493            | 559329.22         | 279777.88         |
| 5  | 2                  | 666                 | 4008            | 561537.09         | 279700.03         |
| 6  | 3                  | 948                 | 4311            | 563743.38         | 279982.05         |
| 7  | 4                  | 1268                | 4479            | 562658.27         | 280044.88         |
| 8  | 5                  | 1637                | 4700            | 568529.52         | 279795.71         |
| 9  | 6                  | 2105                | 4934            | 567359.8          | 279900.81         |
| 10 | 7                  | 2726                | 5421            | 570886.24         | 279891.13         |
| 11 | 8                  | 3616                | 6087            | 582665.15         | 279987.18         |
| 12 | 9                  | 5301                | 7731            | 602551.52         | 279815.44         |
| 13 | 10                 | 13244               | 15566           | 697452.17         | 280008.57         |
| 14 | <b>Grand Total</b> | <b>31855</b>        | <b>60730</b>    | <b>5836712.36</b> | <b>2798903.68</b> |

**Table 2.5:** Summary of 2019 customer behavior by profit decile

We create this summary table in the following manner.

- We make a copy of **Data**, renaming it **Decile analysis (ii)**.

- We need to assign a decile label to each customer, where 1 is assigned to the most profitable customers that account for 10% of the 2019 profit, . . . , and 10 is assigned to the least profitable customers that account for 10% of Madrigal's customer profit in 2019.
- We start by sorting the data by Profit from Largest to Smallest.
- Next we create a cumulative profit column. We enter "Cum profit" in cell E1, =D2 in cell E2, =E2+D3 in cell E3, and copy this second formula down to cell E31856.
- Scrolling up from the bottom of this table, we note that cumulative profit peaks at \$2,802,771.95 (cell E31593) before declining to the total customer profit number of \$2,798,903.68 due to the presence of 263 loss-making customers (=COUNTIF(D2:D31856, "<0")).
- With a total customer profit of \$2,798,903.68, customers whose associated Cum profit number is less than or equal to \$279,890.37 will be assigned to the first decile, customers whose associated Cum profit number is between \$279,890.37 and \$559,780.74 (= \$279,890.37 + \$279,890.37) will be assigned to the second decile, as so on.
- The next step is to determine the individual customer profit numbers associated with these decile thresholds.
- We enter =E31856/10 in cell H2, =H2+H\$2 in cell H3, and copy this formula down to cell H10. These are the boundaries between each decile in terms of cumulative profit.
- In order to compute the individual profit numbers associated with these thresholds, we enter =MAX(IF(E\$2:E\$31856>H2,D\$2:D\$31856)) in cell I2<sup>5</sup> and copying the formula down to cell I10. What does \$545.53 represent? Scrolling down to row 346, we see that it is the Profit number for the first customer where Cum profit is greater than the \$279,890.37 threshold. What about \$345.25? Scrolling down to row 1012, we see that it is the Profit number for the first customer where Cum profit is greater than the \$559,780.74 threshold. And so on.

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<sup>5</sup>This is an array formula; see footnote 2.

- We enter “Decile” in cell F1 and assign a (profit) decile label to each customer by entering  
`=IFS(D2>I$2,1,D2>I$3,2,D2>I$4,3,D2>I$5,4,D2>I$6,5,D2>I$7,6,D2>I$8,7,D2>I$9,8,D2>I$10,9,D2<=I$10,10)`  
in cell F2 and copying the formula down to F31856.
- Table 2.5 is created by inserting a pivot table where Rows is Decile and Values is (Count of) CustomerID, and (Sum of) NumTrans, Spend, and Profit. We rename the resulting worksheet **Decile analysis (Profit)**.

Applying exactly the same steps associated creating Table 2.2 from Table 2.3 to Table 2.5 gives us Table 2.4.<sup>6</sup> Note that we would never present both Tables 2.2 and 2.4. The choice of which one to present will depend on the desired narrative.

### Exercise

Table 2.4 includes in the loss-making customers in the 10th decile. Create a decile table in which the loss-making customers are a separate group.

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<sup>6</sup>Looking at the row associated with the tenth decile, we see that the average margin is much higher than that reported in the equivalent table in *TCBA*. This is simply a byproduct of using a 1% sample of the original data.

# Chapter 3

## Lens 2 Analyses

The primary objective of Lens 2 analyses is to identifying the changes in buyer behavior that underlie period-by-period fluctuations in firm performance. In this chapter we show how to perform all the analyses presented in Chapter 4 of *TCBA*.

### 3.1 Creating the Working Dataset

All the figures and tables that we will create in the course of our analysis of the changes in customer behavior that lie behind the change in overall firm performance from one period to the next can be created from a dataset that reports, for each customer that was active in at least one of the two periods (2018 and 2019), the number of transactions they made in each period, their total spend in each period, and (assuming cost data are available) the profit associated with their purchases in each period. An extract of the required working dataset is given in Figure 3.1.

We can create this dataset in the following manner.

- We open `cust_data_long.csv` in Excel and save it as an `xlsx` file (e.g., `creating_lens_2_data.xlsx`).
- As we will be aggregating the data up to the annual level, we need to identify the year associated with each record. We have a `YearQuarter` variable in column C. To create a `Year` variable, we first enter “Year” in cell G1. Next we enter `=LEFT(C2,5)` in cell G2 and copy this formula down to the last row of the dataset (cell G130790).

|       | A          | B           | C           | D            | E           | F           | G            |
|-------|------------|-------------|-------------|--------------|-------------|-------------|--------------|
| 1     | CustomerID | trans_y2018 | spend_y2018 | profit_y2018 | trans_y2019 | spend_y2019 | profit_y2019 |
| 2     | 1          | 6           | 292.36      | 126.76       |             |             |              |
| 3     | 2          | 3           | 149.62      | 70.97        |             |             |              |
| 4     | 3          | 1           | 35.39       | 9.82         |             |             |              |
| 5     | 4          | 4           | 428.27      | 186.46       | 1           | 58.78       | 30.47        |
| 6     | 6          | 1           | 24.6        | 1.21         |             |             |              |
| 7     | 8          | 11          | 246.25      | 102.84       | 6           | 239.96      | 134.75       |
| 8     | 9          | 3           | 228.54      | 108.97       | 3           | 74.58       | 33.91        |
| 9     | 10         | 16          | 1542.26     | 740.55       | 13          | 1778.24     | 911.73       |
| 10    | 11         | 3           | 324         | 176.7        | 1           | 109.8       | 48.63        |
| 11    | 12         | 3           | 120.91      | 63.18        | 6           | 478.3       | 223.47       |
| 48236 | 70038      |             |             |              | 1           | 131.4       | 68.18        |
| 48237 | 70039      |             |             |              | 1           | 90          | 41.62        |
| 48238 | 70040      |             |             |              | 2           | 413.49      | 216.88       |
| 48239 | 70041      |             |             |              | 1           | 197.4       | 87.35        |

**Figure 3.1:** Working dataset for Lens 2 analyses

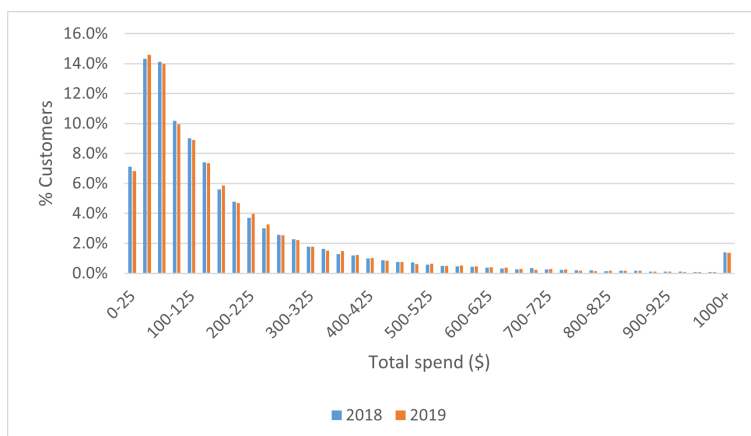
- In order to create the required annual summaries, we insert a pivot table where Rows is CustomerID, Columns is Year, and Values is (Sum of) NumTrans, Spend, and Profit. This creates a worksheet called **Sheet1**. The pivot table contains annual summaries for all four years but we are only interested in the numbers for those customers active in 2018 and/or 2019. Clicking on the Column Labels (cell B3) drop-down filter arrow, we deselect y2016 and y2017.
- We copy cells A5:G48243 into a new worksheet, which we rename **Data**. We then change the column headings (cells A1:G1) to “CustomerID”, “trans\_2018”, “spend\_2018”, “profit\_2018”, “trans\_2019”, “spend\_2019”, and “profit\_2019”. This gives us the dataset reported in Figure 3.1.
- In order to keep our Excel workbook for the analyses presented in this chapter to a manageable size, we copy the **Data** worksheet into a new workbook. (Don’t forget to save this new workbook, say `lens_2_analyses.xlsx`.)
- In the analyses that follow, we will slice and dice this data on the basis of whether a customer was active in both years, just 2018, or just 2019. We create a years variable that indicates each customer’s status. We enter “years” in cell H1, `=IF(B2*E2>0,"both",IF(B2>0,"2018 only","2019 only"))` in cell H2, and copy the formula down to H48239.

## 3.2 Initial Analyses

Summing up the relevant columns in our working dataset, we see that Madrigal’s revenue for 2019 was \$5,836,712, with an associated (customer) profit of \$2,798,904. These numbers represent a 21% and 22% increase, respectively, over Madrigal’s 2018 revenue of \$4,815,502 and (customer) profit of \$2,290,343.

A natural starting point is to repeat the analysis we conducted in Chapter 2 for 2018 and compare the two sets of analyses.

Let us consider the case of spend. Our task is to create Figure 3.2, which plots the relative frequency distribution of spend for 2018 and 2019 on the same set of axes.



**Figure 3.2:** Distribution of total spend in 2018 and 2019

We create this plot in the following manner:

- We will use the same bin cutoffs as used for the distribution of spend across those customers active in 2019. We add a new worksheet, renaming it **Distribution of spend** and enter the bin boundaries. We enter 25 in cell A2, =A2+25 in cell A3, copy this formula down to cell A41, and enter “max” in cell A42.
- We want to count how many customers active in 2018 spent less than or equal to \$25, less than or equal to \$50, and so on. In Chapter 2, we used the COUNTIF function. In this case, we need to add the additional condition that the customer made at least one purchase in 2018. In other words, they are a

“2018 only” or “both” customer; equivalently, they are not a “2019 only” customer. We will make use of Excel’s COUNTIFS function, which applies criteria to cells across multiple ranges and counts the number of times all criteria are met. We enter =COUNTIFS(Data!C\$2:C\$48239, "<="&A2, Data!H\$2:H\$48239, "<>2019 only") in cell B2 and copy this formula down to cell B41. We compute the total number of 2018 buyers by entering =COUNTIF(Data!H2:H48239, "<>2019 only") in cell B42.

- We repeat this for 2019. Noting that the 2019 spend numbers are in column F and that we wish to exclude the “2018 only” customers, we enter =COUNTIFS(Data!F\$2:F\$48239, "<="&A2, Data!H\$2:H\$48239, "<>2018 only") in cell C2 and copy this formula down to cell C41. Finally we enter =COUNTIF(Data!H2:H48239, "<>2018 only") in cell C42.
- The next step is to compute the number of customers falling into each bin. We copy column A to column E and change cell E42 to “>1000”. Next we enter =B2 in cell F2, =B3-B2 in cell F3 and copy this second formula down to cell F42. This gives us the bin frequencies for 2018. We copy cells F2:F42 across to column G to give us the bin frequencies for 2019.
- These two sets of frequencies are quite different. But how much of this difference is simply down to the fact that Madrigal had more customers in 2019 — 26,254 in 2018 (cell B42) versus 31,855 in 2019 (cell C42).
- In order to answer this question, we compute the relative frequencies for each year. We enter =F2/SUM(F\$2:F\$42) in cell H2, copy the formula across and down to cell I42, and format the cells as percentages.

Plotting these relative frequencies gives us Figure 3.2. We observe that the distribution of spend amongst those customers active in 2018 is very similar to that for those customers active in 2019.

The obvious next step is to create similar plots for profit, number of transactions, average spend per transaction, and margin. We leave this as an exercise for the interested reader.

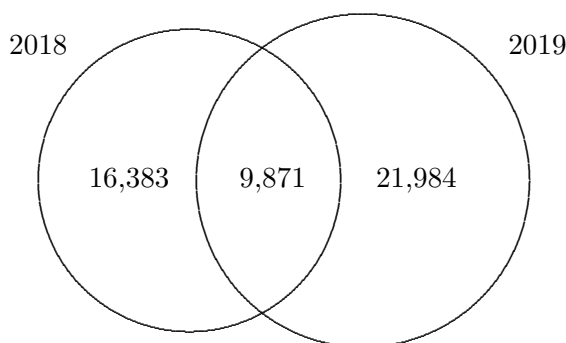
We have seen that Madrigal had 26,254 active customers in 2018 and 31,855 in 2019. How many were active in both years? We answer this question in the following manner. Going back to the **Data**

worksheet, we insert a pivot table where Rows is years and Values is (Count of) CustomerID, resulting in Table 3.1. We see that 9,871 customers were active in both years, which means that 16,383 of the 26,254 customers active in 2018 did not make any purchases in 2019, and 21,984 of the 31,855 customers active in 2019 did not make any purchases in 2018.

| Row Labels         | Count of CustomerID |
|--------------------|---------------------|
| 2018 only          | 16383               |
| 2019 only          | 21984               |
| both               | 9871                |
| <b>Grand Total</b> | <b>48238</b>        |

**Table 3.1:** Number of active customers by year of activity

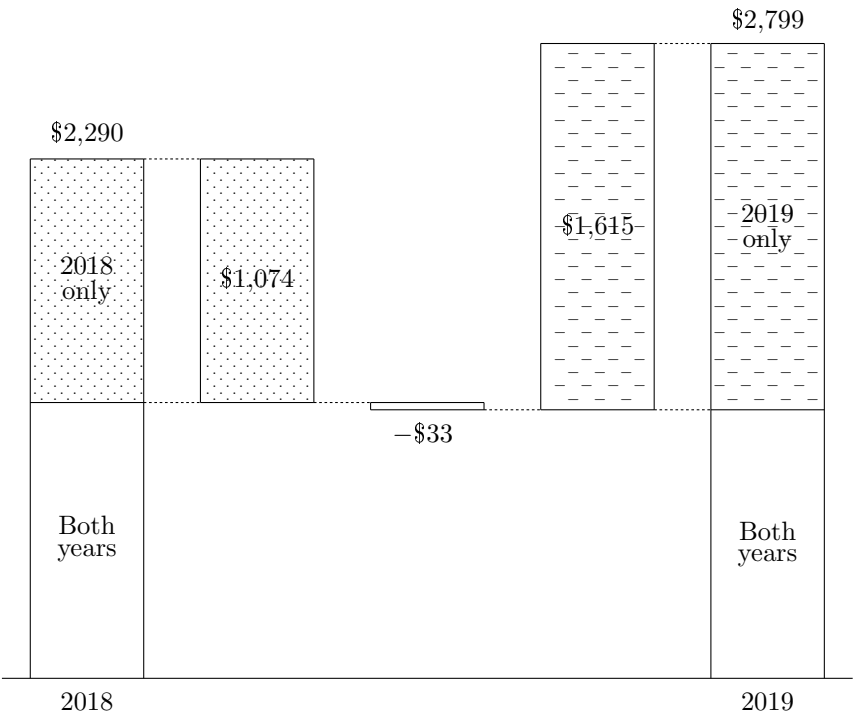
One way of presenting these numbers is via a area-proportional Venn diagram, such as in Figure 3.3. When creating such a figure, the area of each circle should be proportional to the number of customers associated with each year and the area of the overlap of the two circles should be proportional to the number of customers active in each year. At the time of writing, Excel cannot create such plots. In Appendix B we show how to compute the radius of each circle and the distance between the centres of the two circles. Given this information, the area-proportional Venn diagram can be created using a basic drawing package.



**Figure 3.3:** Overlap of the customer base in 2018 and 2019

Customers active in both years represent 38% ( $=9,871/26,254$ ) of

those active in 2018 and 31% ( $=9,871/31,855$ ) of those active in 2019. What proportion of Madrigal's (customer) profits do they account for each year? This is illustrated in Figure 3.4, which we create in the following manner.



**Figure 3.4:** Decomposition of annual (customer) profit (\$000)

Going back to the **Data** worksheet, we insert a pivot table where Rows is years and Values is (Sum of) profit\_y2018 and profit\_y2019. This results in Table 3.2.

| Row Labels  | Sum of profit_y2018 | Sum of profit_y2019 |
|-------------|---------------------|---------------------|
| 2018 only   | 1073945.8           |                     |
| 2019 only   |                     | 1615124.85          |
| both        | 1216396.9           | 1183778.83          |
| Grand Total | 2290342.7           | 2798903.68          |

**Table 3.2:** Profit breakdown by years of activity

We see that the customers active in both years account for 53% of Madrigal's (customer) profit in 2018 and 42% of Madrigal's (customer)

profit in 2019. The profit associated with that group of customers dropped by \$32,618 between the two years. Given these numbers, we create Figure 3.4 using a drawing package.

In order to understand why those customers active in both years represent a greater proportion of the profit than they do of the number of customers active each year, we apply the  $\text{AOF} \times \text{AOV} \times \text{Margin}$  composition we used in the previous chapter to understand inter-decile differences.

The first step is to create a table that reports the number of active customers, total transactions, total spend, and total profit in 2018 and 2019 for the three customer groups (2018 only, 2019 only, both). Going back to the **Data** worksheet, we insert a pivot table where Rows is years and Values is (Count of) CustomerID and (Sum of) trans\_y2018, spend\_y2018, profit\_y2018, trans\_y2019, spend\_y2019, and profit\_y2019. Reformatting the resulting table gives us Table 3.3.

|           |             | 2018        | 2019        |
|-----------|-------------|-------------|-------------|
| 2018 only | # customers | 16,383      |             |
|           | tot trans   | 22,975      |             |
|           | tot spend   | \$2,227,954 |             |
|           | tot profit  | \$1,073,946 |             |
| 2019 only | # customers |             | 21,984      |
|           | tot trans   |             | 33,160      |
|           | tot spend   |             | \$3,346,675 |
|           | tot profit  |             | \$1,615,125 |
| both      | # customers | 9,871       | 9,871       |
|           | tot trans   | 27,813      | 27,570      |
|           | tot spend   | \$2,587,548 | \$2,490,037 |
|           | tot profit  | \$1,216,397 | \$1,183,779 |

**Table 3.3:** Performance summary by customer group

Noting that  $\text{AOF} = \text{total trans} / \text{number of customers}$ ,  $\text{AOV} = \text{total spend} / \text{total trans}$ , and  $\text{Margin} = \text{total profit} / \text{total spend}$ , it is a trivial exercise to create Table 3.4.

### 3.3 Decile Change Analysis

In the previous chapter we introduced the idea of profit deciles to understand how reliant the firm is on a small group of customers. Once we start examining year-on-year performance through the lens of the customer, it is natural to ask about the stability of decile membership.

|           |        | 2018    | 2019     |
|-----------|--------|---------|----------|
| 2018 only | AOF    | 1.40    |          |
|           | AOV    | \$96.97 |          |
|           | Margin | 48%     |          |
| 2019 only | AOF    |         | 1.51     |
|           | AOV    |         | \$100.93 |
|           | Margin |         | 48%      |
| both      | AOF    | 2.82    | 2.79     |
|           | AOV    | \$93.03 | \$90.32  |
|           | Margin | 47%     | 48%      |

**Table 3.4:** Basic temporal decomposition of performance

Are high-value customers in one period still high-value customers in the next period? We can explore this using a decile change analysis. Table 3.5 presents the output of a profit (as opposed to customer) decile change analysis for 2018 and 2019, one in which the decile boundaries remain the same between the two years.

We create this table in the following manner.

- We make a copy of the **Data** worksheet, renaming it **Decile**. We copy and paste the values of cells H2:H48239 onto themselves. Next we delete columns B, C, E, and F (i.e., keep CustomerID, profit\_y2018, profit\_y2019, and year).
- The first thing we need to do is compute the decile boundaries for each year. We did this for those customers active in 2019 in the previous chapter, but let's create the boundaries for both years in one place. We add a new worksheet and rename it **Decile boundaries**. We copy columns A, B, and D (CustomerID, profit\_y2018, and year) from **Data** to columns A:C of **Decile boundaries**, then we copy columns A, C, and D (CustomerID, profit\_y2019, and year) from **Data** to columns F:H of **Decile boundaries**.
- We want to identify the individual customer profit numbers associated with the bins that define 10% of customer profit in 2018. We sort A1:C48239 by profit\_y2018 from Largest to Smallest. We enter “cum profit 2018” in cell D1, =B2 in cell D2, =D2+B3 in cell D3, and copy this second formula down to cell D48239.
- With a total customer profit of \$2,290,342.70, customers whose associated cum profit 2018 number is less than or equal to

|    | A | B | C | D | E | F | G            | H    | I    | J    | K    | L    | M    | N     | O     |       |           |       |              |
|----|---|---|---|---|---|---|--------------|------|------|------|------|------|------|-------|-------|-------|-----------|-------|--------------|
| 1  |   |   |   |   |   |   | 2019 decile  |      |      |      |      |      |      |       |       |       |           |       |              |
| 2  |   |   |   |   |   |   | 1            | 2    | 3    | 4    | 5    | 6    | 7    | 8     | 9     | 10    | 2018 only | Total | % 2018 cust. |
| 3  |   |   |   |   |   |   | 67           | 44   | 29   | 15   | 18   | 15   | 10   | 11    | 7     | 8     | 52        | 276   | 1.1%         |
| 4  |   |   |   |   |   |   | 33           | 63   | 65   | 43   | 45   | 29   | 25   | 29    | 22    | 43    | 150       | 547   | 2.1%         |
| 5  |   |   |   |   |   |   | 19           | 47   | 61   | 71   | 55   | 51   | 39   | 56    | 58    | 65    | 261       | 783   | 3.0%         |
| 6  |   |   |   |   |   |   | 21           | 37   | 53   | 60   | 60   | 63   | 65   | 84    | 74    | 110   | 416       | 1043  | 4.0%         |
| 7  |   |   |   |   |   |   | 23           | 35   | 58   | 75   | 77   | 79   | 81   | 79    | 83    | 146   | 591       | 1327  | 5.1%         |
| 8  |   |   |   |   |   |   | 21           | 40   | 53   | 61   | 98   | 77   | 86   | 106   | 130   | 199   | 924       | 1795  | 6.8%         |
| 9  |   |   |   |   |   |   | 7            | 8    | 30   | 44   | 61   | 68   | 75   | 91    | 128   | 231   | 1150      | 2008  | 7.6%         |
| 10 |   |   |   |   |   |   | 13           | 28   | 61   | 64   | 84   | 98   | 119  | 161   | 197   | 349   | 1932      | 3106  | 11.8%        |
| 11 |   |   |   |   |   |   | 15           | 26   | 48   | 58   | 90   | 129  | 155  | 200   | 272   | 568   | 3035      | 4596  | 17.5%        |
| 12 |   |   |   |   |   |   | 11           | 31   | 58   | 88   | 118  | 181  | 176  | 305   | 477   | 1456  | 7872      | 10773 | 41.0%        |
| 13 |   |   |   |   |   |   | 115          | 252  | 451  | 655  | 1009 | 1442 | 1661 | 2672  | 3951  | 9776  |           | 21984 |              |
| 14 |   |   |   |   |   |   | 346          | 633  | 981  | 1251 | 1722 | 2239 | 2508 | 3831  | 5393  | 12951 | 16383     | 48238 |              |
| 15 |   |   |   |   |   |   | % 2019 cust. | 1.1% | 2.0% | 3.1% | 3.9% | 5.4% | 7.0% | 12.0% | 16.9% | 40.7% |           |       |              |

Table 3.5: Profit decile change (from 2018 to 2019)

\$229,034.27 will be assigned to the first decile, customers whose associated cum profit y2018 number is between \$229,034.27 and \$458,068.54 ( $= \$229,034.27 + \$229,034.27$ ) will be assigned to the second decile, as so on. We want to find the individual customer profit numbers associated with these numbers.

- If you are thinking “What about those 2019 only customers?” We don’t have to worry about them right now. We are simply identifying the boundaries, not assigning individuals to decile bins.
- We enter  $=D48239/10$  in cell K2,  $=K2+K\$2$  in cell K3, and copy this formula down to cell K10. These are the boundaries between each decile in terms of cumulative profit.
- We compute the individual profit numbers associated with these thresholds by entering  $=MAX(IF(D\$2:D\$48239>K2,B\$2:B\$48239))$  in cell L2 and copying the formula down to cell L10. A quick check: What does \$546.28 represent? Scrolling down to row 275, we see that it is the profit\_y2018 number for the first customer where cum profit y2018 is greater than the \$229,034.27 threshold. Etc.
- We repeat this exercise for 2019. We first sort F1:H48239 by profit\_y2019 from Largest to Smallest. Then we enter “cum profit 2019” in cell I1,  $=G2$  in cell I2,  $=I2+G3$  in cell I3, and copy this second formula down to cell I48239.
- We compute the boundaries between each decile in terms of cumulative profit by entering  $=I48239/10$  in cell K13,  $=K13+K\$13$  in cell K14, and copy this formula down to cell K21.
- The individual profit numbers associated with these 2019 thresholds are computed by entering  $=MAX(IF(I\$2:I\$48239>K13,G\$2:G\$48239))$  in cell L13 and copying the formula down to cell L21.

Comparing the decile thresholds for 2018 (cells L2:L10) and 2019 (L13:L21), we see that they are very similar. We will therefore use a common set of cut-offs for both years. We compute the average across the two years and round to the nearest \$5. Entering  $=AVERAGE(L2,L13)$  in cell N2 and copying the formula down to N10,

we choose the following thresholds: 545, 350, 255, 195, 150, 115, 90, 65, and 40.

Note that our use of common thresholds means that each decile will no longer represent exactly 10% of Madrigal's profit for each year. Whether one uses year-specific thresholds or common thresholds is a matter of analyst judgment.

The next step is to assign each customer to a year-specific decile.

- Rather than use the IFS function as in the previous chapter, let us consider an alternative approach that makes use of the VLOOKUP function.
- If you have used VLOOKUP before, it is very likely that you were looking for exact matches. Here we will be assigning customers to decile bins based on which profit range their profit falls into. We will be performing what is called an approximate or non-exact match. There are two things we need to be aware of when performing an approximate match. First, it is assumed that the table being interrogated is sorted in ascending order. Second, the function starts at the top of the table being interrogated and works its way down to find the value we are looking for. If it finds a value that is greater than the lookup value, it goes up one row and presents the previous value as the match result. In other words, it returns the largest value in the table that is less than or equal to lookup value.
- Reflecting on the thresholds identified above, a customer whose profit is less than or equal to \$40 will be assigned to decile 10. If their profit is greater than \$40 and less than or equal to \$65, they will be assigned to decile 9. And so on. In light of the way that VLOOKUP works, we will create a table that contains the lower bound of the range associated with each decile bin.
- Going back to the **Decile** worksheet, we first need to create the lookup table. We enter "Decile" in cell I1, and 10, 9, ..., 1 in cells I2:I11. The lower bound for decile 10 is the minimum profit observed, so we enter =MIN(B1:C48239) in cell H2. If the upper bound for the 10th decile is \$40, the lower bound for the 9th decile will be just above \$40. Working on the assumption that the profit numbers are computed to the nearest cent, we enter 40.001 in cell H3. We continue down column H, adding 0.001 to

the threshold numbers identified above, ending with 545.001 in cell H11.

- The next step is to assign customers to decile. We do this separately for each year. To assign people to deciles for each year, we enter “Decile 18” in cell E1, “Decile 19” in cell F1, `=VLOOKUP(B2,$H$2:$I$11,2)` in cell E2, and copy the formula across and down to F48239.
- Does it matter that we are assigning people who were not active in a year (and therefore have a profit of 0) to a decile? As we shall soon see, the answer is no.
- The next step is to count how many customers in deciles 1–10 in 2018 were in deciles 1–10 in 2019, assuming they were active in both years. We insert a pivot table, where Rows is Decile 18, Columns is Decile 19, Values is (Count of) CustomerID, and we Filters on years, selecting “both”. (We rename the new worksheet **wip 1**.) The resulting table tells us how many of the 9871 customers who were active in both years were in decile  $i$  ( $= 1, \dots, 10$ ) in 2018 and decile  $j$  ( $= 1, \dots, 10$ ) in 2019.
- What about those customers that were only active in 2018? How many are assigned to each decile? Going back to **Decile**, we insert a pivot table where Rows is Decile 18, Values is (Count of) CustomerID, and we Filters on years, selecting “2018 only”. (We rename the new worksheet **wip 2**.) The resulting table tells us how many of the 16,383 customers active only in 2018 were in deciles 1–10.
- What about those customers that were only active in 2019? We go back to **Decile** and insert a pivot table where Columns is Decile 19, Values is (Count of) CustomerID, and we Filters on years, selecting “2019 only”. (We rename the new worksheet **wip 3**.) The resulting table tells us how many of the 21,984 customers active only in 2019 were in deciles 1–10.
- Combining these three tables in a new worksheet (which we rename **Decile change table**) gives us cells C3:M13 of Table 3.5. The row totals in column N are the sum of columns C:M. Similarly, the column totals in row 14 are the sum of rows 3:13.

The percentage in column O are the associated column N numbers divided by 26,254, the number of active customers in 2018. Similarly, the percentages in row 15 are the associated row 14 numbers divided by 31,855, the number of active customers in 2019.

### 3.4 Up-down Analysis

If we wish to dig deeper into the changes in customer-level profitability for those customers active in both years, we can undertake what we call an up-down analysis. For each customer we ask whether their 2019 profit was high or lower than their 2018 profit. In order to understand what lies behind their change in profit, we decompose each customer's annual profit into number of transactions  $\times$  average spend per transaction  $\times$  (average) margin and, for each component, ask whether it increased or decreased from one year to the next. These four up/down questions result in 16 possible groups of customers. We then look at the amount of profit change associated with each group. This results in Table 3.5.

|    | K      | L       | M    | N         | O           | P           | Q           | R            |
|----|--------|---------|------|-----------|-------------|-------------|-------------|--------------|
| 3  |        |         |      |           |             | Profit      |             |              |
| 4  | Profit | # Trans | ASPT | Avg Marg  | # Customers | 2018        | 2019        | Change       |
| 5  | Up     | Up      | Up   | Up        | 1836        | \$103,979   | \$325,405   | \$221,426    |
| 6  | Up     | Up      | Up   | Down      | 1212        | \$77,246    | \$194,192   | \$116,946    |
| 7  | Up     | Up      | Down | Up        | 714         | \$68,717    | \$123,570   | \$54,853     |
| 8  | Up     | Up      | Down | Down      | 510         | \$52,628    | \$93,129    | \$40,501     |
| 9  | Up     | Down    | Up   | Up        | 336         | \$37,361    | \$52,712    | \$15,351     |
| 10 | Up     | Down    | Up   | Down      | 118         | \$15,297    | \$20,408    | \$5,110      |
| 11 | Up     | Down    | Down | Up        | 17          | \$363       | \$693       | \$330        |
| 12 | Down   | Up      | Up   | Down      | 163         | \$8,926     | \$7,396     | -\$1,530     |
| 13 | Down   | Up      | Down | Up        | 812         | \$89,648    | \$56,035    | -\$33,613    |
| 14 | Down   | Up      | Down | Down      | 1277        | \$149,840   | \$81,557    | -\$68,283    |
| 15 | Down   | Down    | Up   | Up        | 529         | \$94,876    | \$57,942    | -\$36,935    |
| 16 | Down   | Down    | Up   | Down      | 485         | \$89,816    | \$48,750    | -\$41,067    |
| 17 | Down   | Down    | Down | Up        | 930         | \$202,938   | \$67,615    | -\$135,323   |
| 18 | Down   | Down    | Down | Down      | 931         | \$224,713   | \$54,382    | -\$170,331   |
| 19 |        |         |      |           | 9870        | \$1,216,349 | \$1,183,785 | -\$32,564    |
| 20 |        |         |      | 2018 only | 16383       | \$1,073,946 |             | -\$1,073,946 |
| 21 |        |         |      | 2019 only | 21984       |             | \$1,615,125 | \$1,615,125  |
| 22 |        |         |      | Total     | 48237       | \$2,290,295 | \$2,798,910 | \$508,615    |

**Figure 3.5:** Up-down analysis

We create this table in the following manner.

- We go back to **Data**, turn on filtering (Ctrl+Shift+L), click on the drop-down filter arrow for years (cell H1) and deselect “2018 only” and “2019 only”. We copy columns A:G of the filtered table into a new worksheet, which we rename **Up-down data**. (There should be 9872 rows in this new worksheet.)
- First, we need to create a variable that indicates whether each customer’s profit increased or decreased from 2018 to 2019. We enter “p\_u-d” in cell H1 (short for profit up-down),  $=1*(G2>=D2)$  in cell H2, and copy this formula down to cell H9872.
- Next, we need to create a variable that indicates whether each customer’s number of transactions increased or decreased from 2018 to 2019. We enter “t\_u-d” in cell I1 (short for transactions up-down),  $=1*(E2>=B2)$  in cell I2, and copy this formula down to cell I9872.
- Next, we need to create a variable that indicates whether each customer’s average spend per transaction increased or decreased from 2018 to 2019. We enter “as\_u-d” in cell J1 (short for average spend up-down),  $=1*(F2/E2>=C2/B2)$  in cell J2, and copy this formula down to cell J9872.
- Finally, we need to create a variable that indicates whether each customer’s average margin increased or decreased from 2018 to 2019. We enter “m\_u-d” in cell K1 (short for margin up-down),  $=1*(G2/F2>=D2/C2)$  in cell K2, and copy this formula down to cell K9872. Don’t worry about the #DIV/0! in cell K10.
- As noted above, these four up/down indicators result in 16 possible groups of customers. We create this group variable using the CONCAT function. We enter “group” in cell L1,  $=CONCAT(H2:K2)$  in cell L2, and copy the formula down to L9872.
- In order to count the number of customers associated with each up-down group, we insert a pivot table where Rows is group and Values is (Count of) CustomerID and (Sum of) profit\_y2018 and profit\_y2019. We rename the new worksheet **Up-down table**.
- The first row in the pivot table is 0000, which corresponds to Down Down Down Down. We would like the table to start with

Up Up Up Up, which corresponds to 1111. Clicking on the drop-down filter/sort arrow for Row Labels (cell A3), we Sort Z to A.

- Note that we have a #DIV/0! group containing one customer. Looking back at row 10 of **Up-down** data, we see that this customer had a total spend of 0, which means their margin is undefined. We will exclude them from our summary table. (Looking closely at Table 3.5, we see that it reports 9,870 “both” customers, one less than the 9871 reported in Tables 3.1 and 3.3 and Figure 3.3. If this worries you, you could always assign this customer to the Down Down Down Down group.
- The next step is to “explode” the group variable back into four columns of 0s and 1s so that we can create the associated “Up” and “Down” labels for the table.
- We start by entering work labels “profit”, “trans”, “avg spend”, and “margin” in cells F4:I4.
- Excel’s MID function allows us to extract a specific number of characters from a text string, starting at the position we specify. In order to extract the first character from the binary string in group (which indicates whether or not the customers associated with this particular pattern saw their profit increase or decrease from 2018 to 2019), we enter =MID(A5,1,1) in cell F5 and copy the formula down to cell F18.
- Did the customers in this pattern group have more or fewer transactions in 2019 compared to 2018? We enter =MID(A5,2,1) in cell G5. Similarly, we enter =MID(A5,3,1) in cell H5 for average spend per transaction and =MID(A5,4,1) in cell I5 for margin. We copy cells G5:I5 down to row 18.
- We are now in a position to create Table 3.5. Having entered the relevant table headings in cells K4:R4, we want to convert the 0s and 1s in cells F5:I18 into “Down” and “Up”. We enter =IF(F5="1","Up","Down") in cell K5 and copy the formula across and down to cell N18. Note the quotation marks around 1.<sup>1</sup> While the 0s and 1s we created in **Up-down data** were

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<sup>1</sup>We could replace “Up” and “Down” with up- and down-arrows by using the

numbers, combining them via the `CONCAT` function created a string and `MID` extracts the individual elements as a string, not a number.

- Next we copy the relevant numbers from the pivot table output. We enter `=B5` in cell O5 and copy the formula across and down to cell Q18.
- We compute the change in the total profit associated with each group by entering `=Q5-P5` in cell R5 and copying the formula down to cell R18.
- We compute the column totals for the group of customers that were active in both years by entering `=SUM(O5:O18)` in cell O19 and copying the formula across to R19.
- The next step is to add the profit numbers for the 2018 only and 2019 only groups. We enter the relevant numbers from Tables 3.1 and 3.2 in cells O20:Q21 and copy the change in profit formula from (say) cell R18 to cells R20:R21.
- Finally, we compute the totals. We enter `=SUM(O19:O21)` in cell O22 and copy the formula across to cell R22.

Looking at the output, we see that there are 14 Up/Down groups, not 16. We see that Up Down Down Down and Down Up Up Up are missing. At first glance, this may seem logical. How could someone's profit go up while their number of transactions, average spend per transaction and margin go down? Actually it is possible if they were loss making in 2018 and they are less unprofitable in 2019. Similarly, Down Up Up Up can occur; an unprofitable customer becomes more unprofitable because they spend more in total (Up Up) and their margin becomes less negative (Up). Such customers are rare but may appear in such analyses. In fact, these groups are present in *TCBA* Figure 4.3. There are 29 Up Down Down Down customers and 38 Down Up Up Up customers in the full dataset. It is not surprising that they are missing in this 1% sample dataset.

As we were creating this table, we labeled someone as Up if their 2019 value on the relevant dimension was greater than or equal to

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formula `=IF(F5="1",UNICHAR(8593),UNICHAR(8595))`. We may also want to color code the cells, green for up and red for down.

2018 value. The likelihood of a customer's profit being exactly the same from one year to the next is rather small. Ditto, to a lesser extent, for their (total) spend. However, it is more likely that their number of transactions is the same between the two years. Going back to **Up-down data**, we can compute the number of customers (active in both years) whose profit was the same in each year using the formula `=SUM(1*(D2:D9872=G2:G9872))`.<sup>2</sup> We see that four of the 4743 (`=SUM(H2:H9872)`) customers labeled Up for profit had the same profit between the two years. Looking at spend, we see that 41 (`=SUM(1*(C2:C9872=F2:F9872))`) customers had the same spend each year, while 5102 (`=SUM(1*(C2:C9872>F2:F9872))`) customers had a higher (total) spend in 2019 compared to 2018.

In contrast, of the 6524 (`=SUM(I2:I9872)`) customers labeled Up for transactions, 3273 (`=SUM(1*(B2:B9872=E2:E9872))`) had the same number of transactions in each year. In other words, one third of the customers active in both years made the same number of transactions in 2019 as they did in 2018. It could therefore be argued that we use three levels (Up Same Down) for transactions when creating Table 3.5.

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<sup>2</sup>Note: This is an array formula.

## Chapter 4

# Lens 3 Analyses

The primary objective of Lens 3 analyses is to explore how customer behavior evolves over time. Central to these analyses is the idea of a cohort, defined as a group of customers acquired in a particular period. In this chapter we show to perform all the analyses presented in Chapter 5 of *TCBA* that can be performed using the “quarterly” data we have at hand. This means we cannot recreate *TCBA*, Figures 5.1 and 5.8, which require transactions to be reported on a weekly or daily basis.

### 4.1 Creating the Working Dataset (Part 1)

The first thing we need to do is extract the records associated with the Q1 2016 cohort.

- We open `cust_data_long.csv` in Excel.
- We turn on filtering (Ctrl+Shift+L), click on the drop-down filter arrow for `Cohort`, and then deselect all the options except “y2016\_q1”. (The easiest way to do this is to untick “Select All” and then select “y2016\_q1”.)
- We copy columns A:F of the filtered table into a worksheet in a new workbook, which we rename **Raw Data**. There should be 7430 rows in this worksheet. (Don’t forget to save this new workbook, say `creating_lens_3_data.xlsx`)

Our initial analyses are based off the data summary given in Figure 4.1, which for each quarter reports the number of cohort members

that were active in that quarter, the total number of transactions they made, and their total spend. We create it in the following manner.

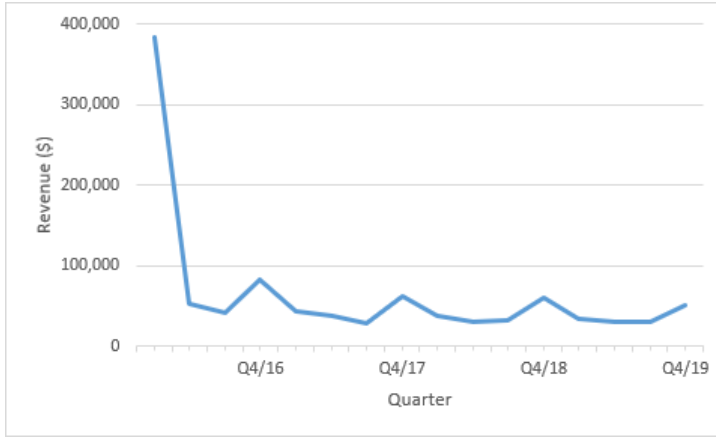
- We insert a pivot table where Rows is YearQuarter and Values is (Count of) CustomerID and (Sum of) NumTrans and Spend, and Profit.
- We copy the pivot table output (ignoring the Grand Total row) and paste into a new worksheet (which we rename **Q1 2016 cohort summary**) in a new workbook, renaming the columns “Quarter”, “NumCust”, “TotTrans”, and “TotSpend”. (Don’t forget to save this new workbook, say `lens_3_analysis.xlsx`)

|    | A        | B       | C        | D        |
|----|----------|---------|----------|----------|
| 1  | Quarter  | NumCust | TotTrans | TotSpend |
| 2  | y2016_q1 | 2944    | 3622     | 383934.4 |
| 3  | y2016_q2 | 355     | 569      | 53165.2  |
| 4  | y2016_q3 | 326     | 489      | 41085.58 |
| 5  | y2016_q4 | 481     | 809      | 82607.04 |
| 6  | y2017_q1 | 299     | 444      | 43107.92 |
| 7  | y2017_q2 | 277     | 455      | 37829.47 |
| 8  | y2017_q3 | 236     | 349      | 28130.53 |
| 9  | y2017_q4 | 372     | 643      | 61873.26 |
| 10 | y2018_q1 | 262     | 376      | 38053.59 |
| 11 | y2018_q2 | 251     | 367      | 31123.97 |
| 12 | y2018_q3 | 248     | 395      | 31523.39 |
| 13 | y2018_q4 | 382     | 665      | 59467.24 |
| 14 | y2019_q1 | 221     | 349      | 33296.54 |
| 15 | y2019_q2 | 216     | 354      | 31024.32 |
| 16 | y2019_q3 | 218     | 353      | 30253.23 |
| 17 | y2019_q4 | 341     | 562      | 51612.37 |

**Figure 4.1:** Quarterly summary of cohort activity

## 4.2 Initial Analysis

Figure 4.2 is a plot of the quarterly revenue (TotSpend in the table above) associated with this cohort in its acquisition quarter and the subsequent 15 quarters. We notice a massive drop in sales the quarter following acquisition. Thereafter, cohort revenue declines slowly, with minor increases each Q4.



**Figure 4.2:** Q1/2016 cohort quarterly revenue

In order to get some insight into what lies behind these revenue fluctuations, we can apply the following decomposition:

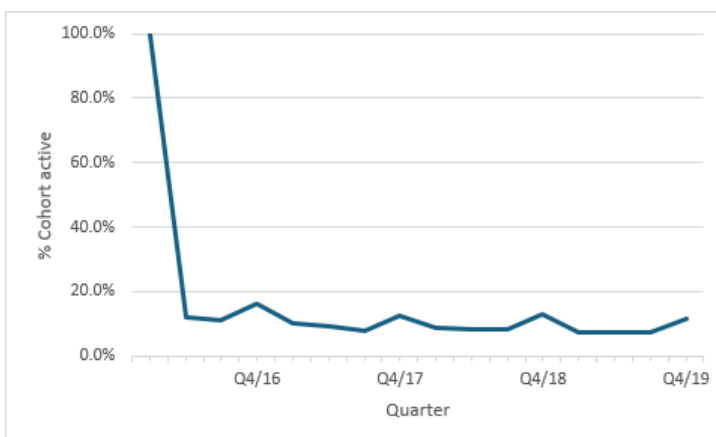
$$\begin{aligned}
 \text{revenue} &= \text{cohort size} \times \frac{\# \text{ active cohort members}}{\text{cohort size}} \\
 &\quad \times \frac{\text{revenue}}{\# \text{ active cohort members}} \\
 &= \text{cohort size} \times \% \text{ cohort active} \\
 &\quad \times \frac{\text{average spend per}}{\text{active cohort member}}
 \end{aligned}$$

We can further decompose average spend per active cohort member:

$$\begin{aligned}
 \frac{\text{revenue}}{\# \text{ active cohort members}} &= \frac{\# \text{ transactions}}{\# \text{ active cohort members}} \\
 &\quad \times \frac{\text{revenue}}{\# \text{ transactions}} \\
 &= \text{average order frequency (AOF)} \times \text{average order value (AOV)}
 \end{aligned}$$

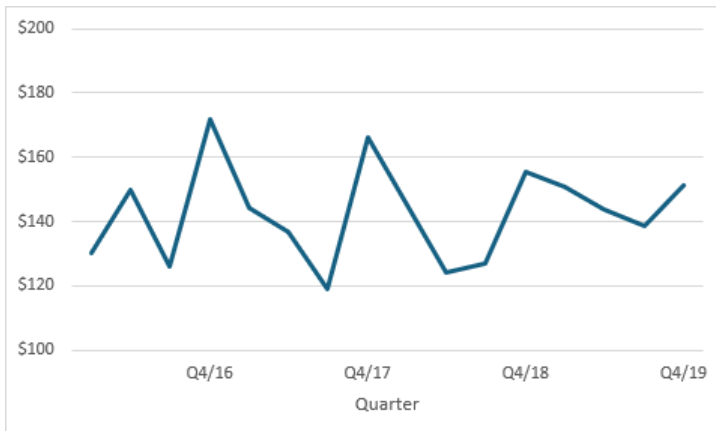
We apply the decomposition in the following manner.

- We compute the % cohort active numbers by entering “% active” in cell F1, =B2/B\$2 in cell F2, formatting the cell as percentage, and then copying the formula down to F17. Plotting these numbers gives us Figure 4.3.



**Figure 4.3:** % Active by quarter

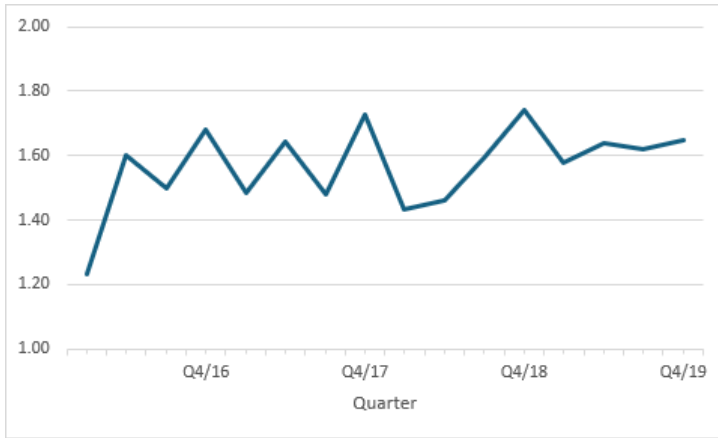
- We compute the average spend per active customer numbers by entering “ASPAC” in cell G1,  $=D2/B2$  in cell G2, formatting the cell as currency, and then copying the formula down to G17. Plotting these numbers gives us Figure 4.4.



**Figure 4.4:** Average spend per active customer by quarter

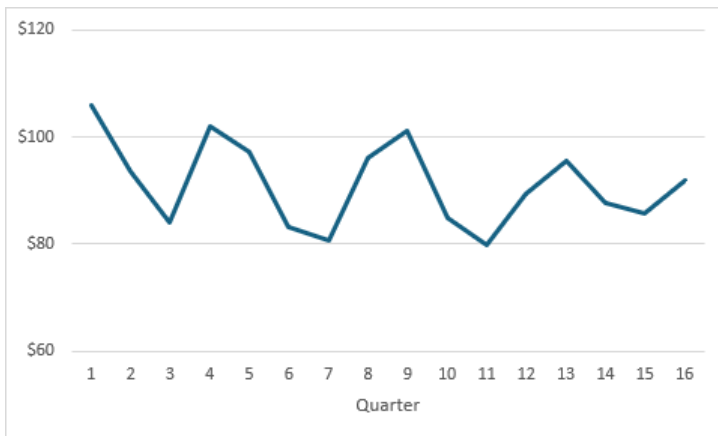
- We compute the AOF numbers by entering “AOF” in cell H1,  $=C2/B2$  in cell H2, and then copying the formula down to H17. Plotting these numbers gives us Figure 4.5.<sup>1</sup>

<sup>1</sup>Comparing this plot (which is based off a 1% sample of the original data) to



**Figure 4.5:** Average order frequency (AOF) by quarter

- We compute the AOV numbers by entering “AOV” in cell I1, =D2/C2 in cell I2, formatting the cell as currency, and then copying the formula down to I17. Plotting these numbers gives us Figure 4.6.



**Figure 4.6:** Average order frequency (AOF) by quarter

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the corresponding AOF plot in *TCBA* (Figure 5.5), we see that the seasonality in AOF is much less pronounced.

## 4.3 Creating the Working Dataset (Part 2)

Our next set of analyses of the behavior of the Q1/2016 cohort requires us to create a new working dataset, one that gives us the total number of transactions made by each customer in each quarter. We create this dataset in the following manner.

- Returning to **Raw data** in `creating_lens_3_data.xlsx`, we insert a pivot table where Rows is CustomerID, Columns is Year-Quarter, and Values is (Sum of) NumTrans.
- We copy the contents of this pivot table (cells A4:Q2948—we do not include the Grand Total row or column) into a new worksheet in `lens_3_analysis.xlsx`, and rename cell A1 “CustomerID”. This gives us the dataset reported in Figure 4.7.
- We rename the worksheet **Data (a)**.

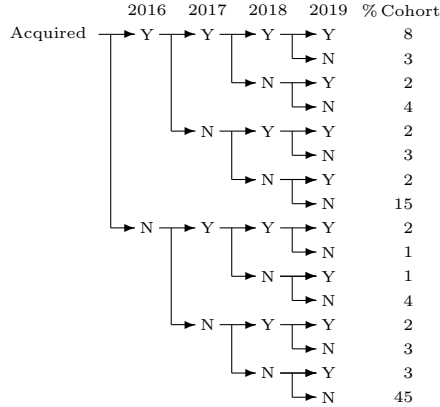
|      | A          | B        | C        | D        | E        | P        | Q        |
|------|------------|----------|----------|----------|----------|----------|----------|
| 1    | CustomerID | y2016_q1 | y2016_q2 | y2016_q3 | y2016_q4 | y2019_q3 | y2019_q4 |
| 2    | 8917       | 5        |          | 2        |          | 4        | 3        |
| 3    | 8918       | 1        | 1        | 1        | 1        | 2        | 2        |
| 4    | 8919       | 1        |          |          |          |          |          |
| 5    | 8920       | 1        | 1        |          |          |          |          |
| 6    | 8921       | 1        |          |          | 1        |          |          |
| 7    | 8922       | 1        |          | 1        |          |          |          |
| 8    | 8923       | 3        |          |          |          |          |          |
| ...  | ...        | ...      | ...      | ...      | ...      | ...      | ...      |
| 2940 | 11855      | 1        |          |          |          |          |          |
| 2941 | 11856      | 1        |          |          |          |          |          |
| 2942 | 11857      | 1        |          |          | 1        |          |          |
| 2943 | 11858      | 1        |          |          |          |          |          |
| 2944 | 11859      | 1        |          |          |          |          |          |
| 2945 | 11860      | 1        |          |          |          |          |          |

**Figure 4.7:** Working data set for understanding the evolution of purchase incidence

## 4.4 Understanding the Evolution of Purchase Incidence

Our first task is to create Figure 4.8.

We perform this analysis in the following manner.



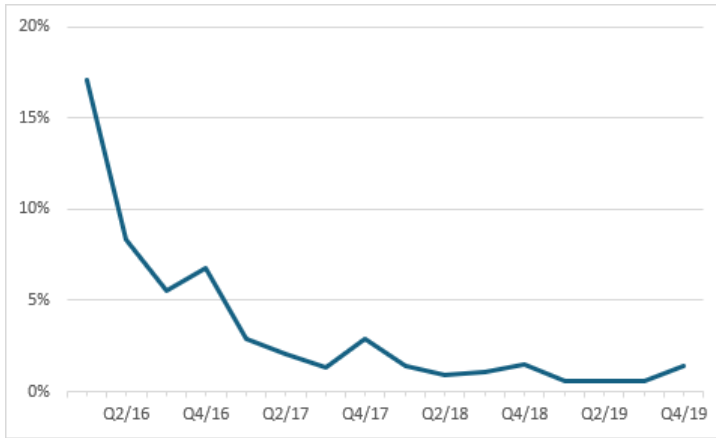
**Figure 4.8:** Patterns of annual buying for the Q1/2016 cohort

- We add a new worksheet, renaming it **Buying patterns (i)**, and copy the contents of the CustomerID column from **Data (a)** into column A.
- We enter the labels “2016”, “2017”, “2018”, and “2019” into cells B1:E1.
- For each member of the cohort, we wish to determine whether or not they made any additional purchases in 2016 (beyond the purchase that signaled their “birth” as a customer). We enter `=1*(SUM('Data (a)'!B2:E2)>1)` in cell B2 and copy this formula down to cell B2945.
- Did the customer make at least one purchase in 2017? We enter `=1*(SUM('Data (a)'!F2:I2)>0)` in cell C2. Similarly, we enter `=1*(SUM('Data (a)'!J2:M2)>0)` in cell D2 for 2018 and `=1*(SUM('Data (a)'!N2:Q2)>0)` in cell E2 for 2019. We copy cells C2:E2 down to row 2945.
- We create a “pattern” variable for each customer using the CONCAT function. We enter “Pattern” in cell F1, `=CONCAT(B2:E2)` in cell F2, and copy the formula down to F2945. (This four-digit binary number captures each customer’s annual repeat-buying pattern.)
- In order to count the number of customers associated with each annual buying pattern, we insert a pivot table where Rows is

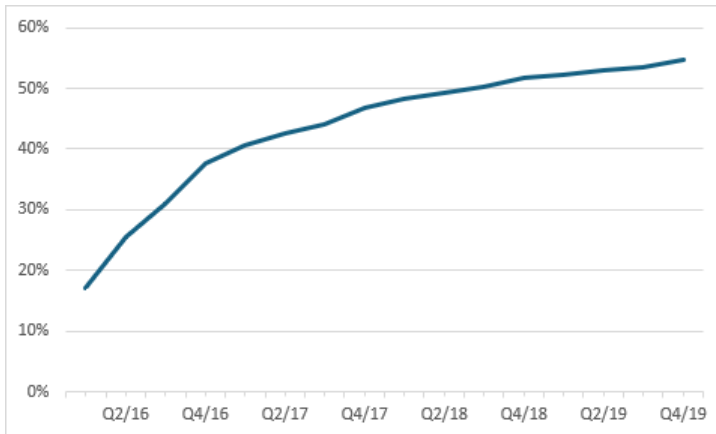
Pattern and Values is (Count of) CustomerID. We rename the new worksheet **Buying patterns (ii)**.

- We copy and paste the values of cells A4:B19 to D4:E19 and sort this block of cells by “Column D” in Order “Z to A”.
- The next step is to “explode” the Pattern variable back into four columns of 0s and 1s. We start by entering the labels “2016”, “2017”, “2018”, and “2019” into cells G3:J3.
- Recall that Excel’s MID function allows us to extract a specific number of characters from a text string, starting at the position we specify. In order to extract the first character from the binary string in Pattern (which indicates whether or not the customers associated with this particular pattern made at least one additional purchase in their year of acquisition), we enter =MID(D4,1,1) in cell G4 and copy the formula down to cell G19.
- Did the customers in this pattern group make a purchase in 2017? We enter =MID(D4,2,1) in cell H4. Similarly, we enter =MID(D4,3,1) in cell I4 for 2018 and =MID(D4,4,1) in cell J4 for 2019. We copy cells H4:J4 down to row 19.
- We compute the percentage of cohort members associated with each buying pattern by entering =E4/SUM(E\$4:E\$19) in cell K4 and copying the formula down to cell K19. (We format the cells as percentage.)
- If presenting this analysis in tabular form, most audiences will find it easier to process Ys and Ns, as opposed to 1s and 0s. We can create such a table by entering =IF(G4="1","Y","N") in cell M4—note the " " around the 1—and copying the formula across and down to cell P19. We then copy the contents of K4:K19 across to Q4:Q19.
- If you want to present this analysis using a diagram such as that given in Figure 4.8, you will need to use a drawing package.

We note that 45% of those customers acquired in Q1/2016 had not made a second purchase by the end of 2019. Figures 4.9 and 4.10 document when the 55% that did make a second purchase did so. We create these figures in the following manner.



**Figure 4.9:** Percentage of cohort members making a second purchase by quarter



**Figure 4.10:** Cumulative percentage of cohort members making a second purchase

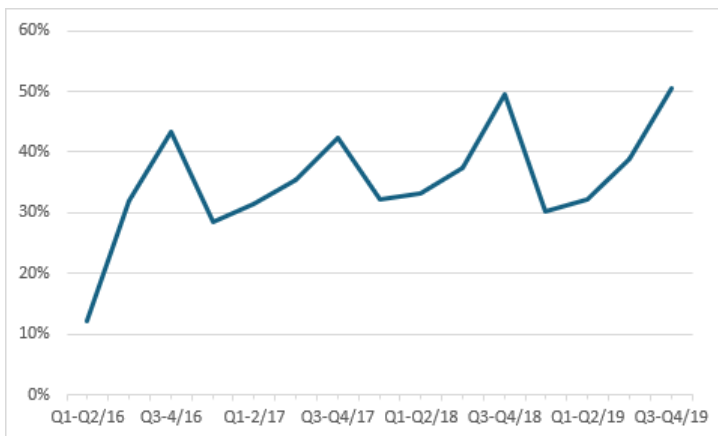
- We make a copy of **Data (a)**, rename the worksheet **Second purchase**, and delete cells B2:Q2945.
- We want to create a table where, reading across the row associated with each customer, a table entry of 0 indicates that they have yet to make a second purchase and 1 indicates that they have made a second purchase (either in the current period or in a previous period).

- Did a customer make a second-ever purchase (aka first-repeat purchase) in the same quarter in which they were acquired? If so, that customer's number of transactions in Q1 2016 is greater than 1. We enter `=1*('Data (a)'!B2>1)` in cell B2 and copy the formula down to cell B2945. An entry of 1 indicates that the customer made at least one additional purchase in Q1 2016, while an entry of 0 indicates that the customer made only one purchase in the quarter in which they were acquired.
- Moving to column C, we want each cell entry to take on a value of 1 if the value of the cell next to it is 1 (i.e., the customer has already made a second purchase), a value of 1 the value in column B was 0 and the customer made at least one purchase in the period associated with column C (i.e., Q2 2016), and a value of 0 if a second-ever purchase was not made in the previous period and the customer did not make any purchases in this quarter. We enter `=MAX(B2,1*('Data (a)'!C2>0))` in cell C2 and copy the formula down to cell C2945.
- We copy cells C2:C2945 across to column Q.
- The next step is to compute how many members of the cohort have made a second purchase by the end of Q1 2016, the end of Q2 2016, ..., the end of Q4 2019. These numbers are the sum of the columns of 0s and 1s in B2:Q2945. We copy the quarter labels from B1:Q1 to T1:AI1. Next we enter `=SUM(B2:Q2945)` in cell T2 and copy the formula across to cell AI2.
- We convert these numbers to percentages by dividing them by the size of the cohort. We enter `=T2/COUNT(A2:A2945)` in cell T4, copy the formula across to cell AI4, and format these cells as percentages. Plotting these numbers gives us Figure 4.10.
- We create the quarter-by-quarter numbers by converting these cumulative percentages to incremental. We enter `=T4` in cell T5, `=U4-T4` in cell U5, and copy this second formula across to cell AI5. Plotting these numbers gives us Figure 4.9.

#### 4.4.1 [Optional] Computing Repeat-Buying Rates

The idea of a repeat-buying rate, also called the repeat rate, is introduced in Chapter 5 of *TCBA* (p. 79) and examined in greater depth

in Chapter 7. This common measure of customer loyalty is defined as the percentage of customers from one period that make at least one purchase in the following period. In this section, we consider how to compute the quarterly repeat-buying rates for the Q1/2016 cohort, an analysis not reported in the book. The quarter-by-quarter repeat-buying rates for this cohort are reported in Figure 4.11, which we can create in the following manner.



**Figure 4.11:** Quarterly repeat-buying rates

- We make a copy of **Data (a)**, rename the worksheet **Repeat buying rate**, and delete cells B2:Q2945.
- We want to create a table where an entry of 0 indicates that a customer made no purchase in that quarter and 1 indicates that a customer has made at least one purchase. We enter `=1*( 'Data (a)' !B2>0)` in cell B2 and copy the formula across and down to cell Q2945.
- Recall that the repeat-buying rate is the percentage of a firm's customers in a given period that also purchase from the firm in the following period. How many customers had at least one transaction with Madrigal in Q1 2016? It is simply the sum of the 0s and 1s in column B. (This is the denominator.) If a customer had at least one transaction in Q1 and at least one transaction in Q2, they will have a 1 in columns B and C. If we multiply these two columns of numbers together, only those

customers that purchased from Madrigal in both periods will have a 1. Any other combination of purchasing will result in a zero. Summing up this product gives us the number of customers that transacted with Madrigal at least once in both quarters. (This is the numerator.)

- We compute the Q1/2016-Q2/2016 repeat buying rate by entering `=SUMPRODUCT(C2:C2945,B2:B2945)/SUM(B2:B2945)` in cell S2. We copy the formula across to cell AG2 and format these cells as percentages.
- Plotting these numbers gives us Figure 4.11.

Note that the repeat-buying rate is a period-by-period number. We are ignoring any repeat purchases made by a customer in Q1/2016. We could have a customer who made five repeat purchases in Q1/2016 and didn't make any other purchases up to the end of 2019. While their repeat purchasing would be captured in Figures 4.9 and 4.10, they would not “appear” as having made a repeat purchase in Figure 4.11.

## 4.5 Creating the Working Dataset (Part 3)

Our next set of analyses see us exploring the cohort members' (undiscounted) profitability over the four year period 2016–2019, which we call value to date (VTD). Our initial analyses require a dataset of the form given in Figure 4.12, which reports the number of transaction, total spend and total profit for each of the 2944 members of the Q1/2016 cohort over the four-year period Q1/2016–Q4/2019.

We create this dataset in the following manner.

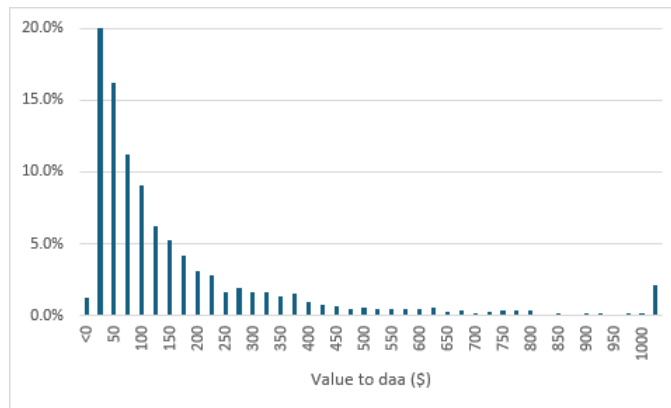
- Returning to **Raw data** in `creating_lens_3_data.xlsx`, we insert a pivot table where Rows is CustomerID and Values is (Sum of) NumTrans, Spend, and Profit.
- We copy the contents of this pivot table (cells A3:Q2947) into a new worksheet in `lens_3_analysis.xlsx`, and rename the column headings (cells A1:A4) “CustomerID”, “SumNumTrans”, “SumSpend”, and “SumProfit”.
- Finally, we rename the worksheet **Data (b)**.

|      | A          | B           | C        | D         |
|------|------------|-------------|----------|-----------|
| 1    | CustomerID | SumNumTrans | SumSpend | SumProfit |
| 2    | 8917       | 29          | 1923.65  | 756.51    |
| 3    | 8918       | 13          | 1126.33  | 550.03    |
| 4    | 8919       | 1           | 26.99    | 7.62      |
| 5    | 8920       | 5           | 624.24   | 228.88    |
| 6    | 8921       | 4           | 105.93   | 44.54     |
| 7    | 8922       | 2           | 167.01   | 51.75     |
| 8    | 8923       | 3           | 394.04   | 128.84    |
| 9    | 8924       | 2           | 204.6    | 94.22     |
| 10   | 8925       | 2           | 98.2     | -4.62     |
| 2941 | 11856      | 1           | 131.94   | 38.71     |
| 2942 | 11857      | 2           | 275.39   | 158.89    |
| 2943 | 11858      | 1           | 31.2     | 12.53     |
| 2944 | 11859      | 1           | 132.19   | 61.3      |
| 2945 | 11860      | 1           | 180      | 115.69    |

**Figure 4.12:** Working dataset for the analysis of VTD

## 4.6 The Distribution of VTD

Our task is to create Figure 4.13, which plots the distribution of VTD across the 2944 members of the Q1/2016 cohort.



**Figure 4.13:** Distribution of Q1/2016 cohort value

The creation of this plot follows the general steps used in Chapter 2. The specifics are as follows.

First, we need to choose our binning strategy and so we start by looking at some descriptive statistics.

- We make a copy of **Data (b)** and rename the copy **Distribution of VTD**. We delete columns B and C, and rename cell B1 “VTD”.
- We compute some basic descriptive statistics.
  - The minimum total spend is computed in cell E2:  
=MIN(B2:B2945)
  - The maximum total spend is computed in cell E3:  
=MAX(B2:B2945)
  - The average total spend is computed in cell E4:  
=AVERAGE(B2:B2945)
  - The median total spend is computed in cell E5:  
=MEDIAN(B2:B2945)
- We see that cohort members’ VTD ranges from –\$23 to \$3756. The mean (\$170) is larger than the median (\$78), with 72% of the cohort members having below-average VTD: =COUNTIF(B2:B2945, "<"&E4)/COUNT(B2:B2945)
- To get further insight into the distribution of VTD across cohort members, we determine the VTD numbers associated with the various percentile levels.
  - We enter 0.05 in cell D8, =D8+0.05 in cell D9, and copy this formula down to D26. Cells D8:D26 are then formatted as percentages.
  - We enter =PERCENTILE.INC(B\$2:B\$2945,D8) in cell E8 and copy this formula down to E26.
- We see that 5% of customers have a VTD at or below \$5.56. 10% have a VTD at or below \$12.05, and so on. A four-year profit greater than \$663.09 puts the cohort member in the top 5% of the cohort.
- We note that just over 2% of the cohort members have a VTD greater than \$1000 (=COUNTIF(B2:B2945, ">1000")/COUNT(B2:B2945)).

Keeping in mind the fact that a narrow bin width implies more bins, we will use a \$25 bin width with \$1000 as the cut-off point for the final bin.

The next step is to determine how many cohort members fall into each bin.

- Staying in **Distribution of VTD**, we first create the bin boundaries. We enter 0 in cell G2, =G2+25 in cell G3, and copy this formula down to cell G42. The upper limit is entered in cell G43: =MAX(B2:B2945).
- For each boundary point, we want to compute the number of customers whose profit was at or below that number. We enter =COUNTIF(B\$2:B\$2945,"<="&G2) in cell H2 and copy the formula down to cell H43.
- In order to compute the number of customers falling into each bin, we first enter =H2 in cell I2. Next we enter =H3-H2 in cell I3 and copy the formula down to cell I43.
- We convert these counts into relative frequencies by entering =I2/SUM(I\$2:I\$43) in cell J2 and copying the formula down to cell J43. Cells J2:J43 are then formatted as percentages.

Plotting these relative frequencies gives us Figure 4.13. We see that there is a lot of variability in VTD. Why are low-value customers low value, and why are high-value customers high value? To get some initial answers to such questions, we perform a VTD-based decile analysis. Our goal is to create Table 4.1, which is a decile analysis where each decile represents 10% of the cohort's VTD.

| Decile | % Cohort | % Trans | % Spend | % VTD | Avg spend/cust | Avg VTD/cust | AOF  | AOV      | Avg margin |
|--------|----------|---------|---------|-------|----------------|--------------|------|----------|------------|
| 1      | 1%       | 8%      | 9%      | 10%   | \$4,079.19     | \$2,073.50   | 35.9 | \$113.71 | 51%        |
| 2      | 2%       | 7%      | 10%     | 10%   | \$2,223.59     | \$1,111.99   | 17.2 | \$129.61 | 50%        |
| 3      | 2%       | 8%      | 10%     | 10%   | \$1,639.46     | \$815.40     | 14.1 | \$116.56 | 50%        |
| 4      | 3%       | 9%      | 10%     | 10%   | \$1,252.80     | \$613.24     | 11.7 | \$107.35 | 49%        |
| 5      | 4%       | 10%     | 10%     | 10%   | \$899.93       | \$441.64     | 9.1  | \$98.54  | 49%        |
| 6      | 5%       | 9%      | 10%     | 10%   | \$677.27       | \$331.06     | 6.3  | \$107.99 | 49%        |
| 7      | 7%       | 9%      | 10%     | 10%   | \$498.15       | \$243.11     | 4.9  | \$101.30 | 49%        |
| 8      | 10%      | 10%     | 10%     | 10%   | \$344.81       | \$167.09     | 3.7  | \$93.55  | 48%        |
| 9      | 16%      | 10%     | 10%     | 10%   | \$213.81       | \$105.52     | 2.3  | \$93.15  | 49%        |
| 10     | 51%      | 20%     | 12%     | 10%   | \$84.87        | \$33.61      | 1.5  | \$58.21  | 40%        |
|        |          |         |         |       | \$352.61       | \$169.78     | 3.7  | \$96.11  | 48%        |

**Table 4.1:** Summary of cohort behavior by VTD decile

The key intermediate step is to create Table 4.2, which reports, for each decile, the number of cohort members in each decile and the

| Row Labels         | Count of CustomerID | Sum of SumNumTrans | Sum of SumSpend   | Sum of SumProfit |
|--------------------|---------------------|--------------------|-------------------|------------------|
| 1                  | 24                  | 861                | 97900.62          | 49763.93         |
| 2                  | 45                  | 772                | 100061.33         | 50039.44         |
| 3                  | 61                  | 858                | 100007.02         | 49739.43         |
| 4                  | 82                  | 957                | 102729.25         | 50285.67         |
| 5                  | 113                 | 1032               | 101692.09         | 49905.45         |
| 6                  | 151                 | 947                | 102267.5          | 49989.41         |
| 7                  | 206                 | 1013               | 102619.56         | 50080.47         |
| 8                  | 299                 | 1102               | 103097.1          | 49958.75         |
| 9                  | 474                 | 1088               | 101345.91         | 50018.23         |
| 10                 | 1489                | 2171               | 126367.63         | 50040.2          |
| <b>Grand Total</b> | <b>2944</b>         | <b>10801</b>       | <b>1038088.01</b> | <b>499820.98</b> |

**Table 4.2:** Source data for VTD decile table

total number of transactions, total spend, and total VTD associated with each decile.

This summary table is created in the following manner.

- We make a copy of **Data (b)**, renaming it **Decile analysis**.
- We need to assign a decile label to each customer, where 1 is assigned to those cohort members with the highest VTD that account for 10% of the cohort's VTD, ..., and 10 is assigned to those cohort members with the lowest VTD that account for 10% of the cohort's overall VTD.
- We sort the data by SumProfit from Largest to Smallest.
- Next we create a cumulative VTD column. We enter "Cum VTD" in cell E1, =D2 in cell E2, =E2+D3 in cell E3, and copy this second formula down to cell E2945.
- With a total cohort VTD of \$499,820.98 (cell E2945), customers whose corresponding Cum VTD number is less than or equal to \$49,982.10 will be assigned to the first decile, customers whose corresponding Cum VTD number is between \$49,982.10 and \$99,964.20 ( $= 49,982.10 + 49,982.10$ ) will be assigned to the second decile, and so on.
- We enter =E2945/10 in cell H2, =H2+H\$2 in cell H3, and copy this formula down to cell H10. These are the boundaries between each decile in terms of cumulative profit.
- We compute the individual profit numbers associated with each of these thresholds by entering =MAX(IF(E\$2:E\$2945>H2,D\$2:

D\$2945)) in cell I2 and copying the formula down to cell I10. What does \$1,373.43 represent? Looking at row 26, we see that this is the SumProfit number for the first cohort member where Cum VTD is greater than the 49,982.10 threshold. And so on.

- We enter “Decile” in cell F1 and assign a (profit) decile label to each customer by entering

```
=IFS(D2>I$2,1,D2>I$3,2,D2>I$4,3,D2>I$5,4,
D2>I$6,5,D2>I$7,6,D2>I$8,7,D2>I$9,8,D2>I$10,9,
D2<=I$10,10)
```

in cell F2 and copying the formula down to F2945.

- Table 4.2 is created by inserting a pivot table where Rows is Decile and Values is (Count of) CustomerID, and (Sum of) NumTrans, Spend, and VTD. (We rename the resulting worksheet **Decile analysis (VTD)**).

Applying the exact same steps associated with the creation of Table 2.3 to Table 4.2 gives us Table 4.1.

Reflecting on Table 4.1, we see that while the higher value deciles have higher AOF and AOV than the less valuable deciles, there is much greater variation in AOF across deciles—decile 1 is almost 25 times higher than decile 10—than we do in AOV, where decile 1 is just under two times higher than decile 10. How is this difference in AOF reflected in customer behavior over time?

In order to explore this, we create Table 4.3, which reports the percentage of the cohort members associated with each decile that are active in each of the four calendar years.

We create this table in the following manner.

- We make a copy of **Decile analysis**, renaming it **Decile pct active (i)**. We copy and paste the values of F1:F2945 to column B and delete contents of columns C: I. We sort the contents of columns A and B by CustomerID from “Smallest to Largest”.
- Next we copy the contents of cells B1:E2945 in **Buying patterns (i)** to cells C1:F2945 in **Decile pct active (i)** (pasting values).
- 2016 currently indicates whether each cohort member made a repeat purchase in 2016. We want this to indicate whether they

| Decile | % Cohort | Pct active 16 | Pct active 17 | Pct active 18 | Pct active 19 |
|--------|----------|---------------|---------------|---------------|---------------|
| 1      | 1%       | 100%          | 100%          | 92%           | 92%           |
| 2      | 2%       | 100%          | 91%           | 84%           | 78%           |
| 3      | 2%       | 100%          | 87%           | 84%           | 87%           |
| 4      | 3%       | 100%          | 83%           | 85%           | 74%           |
| 5      | 4%       | 100%          | 65%           | 73%           | 53%           |
| 6      | 5%       | 100%          | 58%           | 51%           | 50%           |
| 7      | 7%       | 100%          | 45%           | 44%           | 39%           |
| 8      | 10%      | 100%          | 35%           | 34%           | 29%           |
| 9      | 16%      | 100%          | 20%           | 17%           | 16%           |
| 10     | 51%      | 100%          | 7%            | 6%            | 5%            |
|        |          | 100%          | 26%           | 24%           | 21%           |

**Table 4.3:** Annual % active by VTD decile

were active in 2016. By definition, they were. We enter 1 in cell C2 and copy the contents of the cell down to C2945.

- We insert a pivot table where Rows is Decile, and Values is (Count of) CustomerID and (Sum of) 2016, 2017, 2018, and 2019. We rename the new worksheet **Decile pct active (ii)**.
- We enter the decile labels (1–10) in cells H4:H13.
- We compute the percentage of the cohort associated with each decile by entering `=B4/B$14` in cell I4 and copying the formula down to I13.
- We compute the percentage of each decile that is active each year by entering `=C4/$B4` in cell J4 and copying the formula across and down to cell M13. We format these cells as percentage.
- Copying cells J13:M13 down one row gives us the cohort-level annual % active numbers.

## 4.7 RFM Analysis

An RFM analysis “takes stock” of a cohort’s behavior at a given point in time. Our task is to create Table 4.4, which provides such an analysis where we have four “recency” bins, four “frequency” bins, and four “monetary value” bins. In this analysis, we define monetary value as average profit per transaction over the four-year period 2016–2019.

We create this table in the following manner.

|         |                  | Frequency |     |      |     |
|---------|------------------|-----------|-----|------|-----|
| Recency | Avg profit/trans | 1         | 2-4 | 5-10 | 11+ |
| Q1      | < \$25           | 531       | 59  | 2    |     |
|         | \$25-50          | 348       | 61  |      |     |
|         | \$50-75          | 160       | 33  |      |     |
|         | \$75+            | 291       | 33  |      |     |
| Q2-Q8   | < \$25           |           | 123 | 16   | 2   |
|         | \$25-50          |           | 143 | 28   |     |
|         | \$50-75          |           | 74  | 16   | 1   |
|         | \$75+            |           | 90  | 10   | 2   |
| Q9-Q15  | < \$25           |           | 71  | 54   | 18  |
|         | \$25-50          |           | 111 | 81   | 30  |
|         | \$50-75          |           | 52  | 56   | 14  |
|         | \$75+            |           | 53  | 29   | 11  |
| Q16     | < \$25           |           | 18  | 26   | 29  |
|         | \$25-50          |           | 33  | 51   | 54  |
|         | \$50-75          |           | 20  | 23   | 37  |
|         | \$75+            |           | 13  | 28   | 9   |

**Table 4.4:** RFM summary of the Q1/2016 cohort

- We make a copy of **Data (b)** and rename the worksheet **RFM data**.
- The first step is to create the recency variable, which in this case the quarter in which each cohort member made their last transaction.
  - We will label Q1 2016 as 1, Q2 2016 as 2, ..., Q4 2019 as 16.
  - We copy the contents of column A to column L. Next we enter 1, 2, ..., 16 in cells M1:AB1.
  - We enter `=M$1*('Data (a)'!B2>0)` in cell M2 and copy the formula across and down to cell AB2945.
  - We enter “R value” in cell E1, enter `=MAX(M2:AB2)` in cell E2 and copy this formula down to cell E2945.
- Each cohort member’s F value is simply the total number of transactions they made across the four years. We enter “F value” in cell F1, `=B2` in cell F2, and copy this formula down to cell F2945.
- Each cohort member’s M value is their average profit per transaction. We enter “M value” in cell G1, `=D2/B2` in cell G2, and copy this formula down to cell G2945.

- We now create RFM bins:
  - We first segment the members of the cohort on the basis of recency: did they make their last purchase in Q1 (Q1/2016), Q2–Q8 (Q2/2016–Q4/2017), Q9–Q15 (Q1/2018–Q3/2019), or Q4/2019? We enter “R” in cell H1, `=IFS(E2=1,1,E2<=8,2,E2<=15,3,E2=16,4)` in cell H2, and copy this formula down to cell H2945.
  - Next we segment the members of the cohort on the basis of frequency: did they make just one purchase in the past four years, 2–4 purchases, 5–10 purchases, or 11 or more purchases? We enter “F” in cell I1, `=IFS(F2=1,1,F2<=4,2,F2<=10,3,F2>=11,4)` in cell I2, and copy this formula down to cell I2945.
  - Finally, we segment the members of the cohort on the basis of their average profit per transaction: was it less than \$25, between \$25 and \$50, between \$50 and \$75, or greater than \$75? We enter “M” in cell J1, `=IFS(G2<=25,1,G2<=50,2,G2<=75,3,G2>75,4)` in cell J2, and copy this formula down to cell J2945.
- With four recency categories, four frequency categories, and four monetary value categories, there should be 64 possible RFM segments. However, in this application, there are only 52 feasible combinations, as anyone with a frequency of 1 cannot have made a purchase any later than Q1 2016. We count how many cohort members fall into each of the 52 feasible combinations by inserting a pivot table where Rows is R and M, columns is F, and Values is (Count of) CustomerID.

Tidying up the resulting pivot table output and adding the appropriate labels gives us Table 4.4.

Note that there is nothing magical about our choice of four bins for each of R, F, and M and associated bin boundaries. The choices should be made so as to maximize the insight for the recipient of this analysis. Having said that, we feel that it makes sense to have stand-alone recency bins for the first and last quarters. Similarly, it makes sense to have a frequency = 1 bin.

## Chapter 5

# Lens 4 Analyses

Lens 4 sees us extending the decomposition of one cohort’s revenue (as developed as part of our Lens 3 analyses) to profit and using it to compare and contrast the performance of different cohorts. In this chapter we show how to perform all the analyses presented in Chapter 6 of *TCBA* that can be performed using the “quarterly” data we have at hand. This means we cannot recreate *TCBA*, Figures 6.14–6.17, which assume we have access to more temporally disaggregated data.

As a reminder, we can decompose the profit associated with a given cohort in a particular time period using the following multiplicative decomposition:

$$\begin{aligned}\text{cohort profit} &= \text{cohort size} \times \frac{\# \text{ active cohort members}}{\text{cohort size}} \\ &\quad \times \frac{\text{cohort transactions}}{\# \text{ active cohort members}} \\ &\quad \times \frac{\text{cohort revenue}}{\text{cohort transactions}} \\ &\quad \times \frac{\text{cohort profit}}{\text{cohort revenue}} \\ &= \text{cohort size} \times \% \text{ cohort active} \\ &\quad \times \text{Average order frequency (AOF)} \\ &\quad \times \text{Average order value (AOV)} \\ &\quad \times \text{Average margin.}\end{aligned}$$

## 5.1 Creating the Working Dataset

The first step is to create the equivalent of Table 4.1 for all cohorts. We can create this dataset in the following manner.

- We open `cust_data_long.csv` in Excel and save it as an `xlsx` file (e.g., `creating_lens_4_data.xlsx`).
- We insert a pivot table where Rows is Cohort, Columns is Year-Quarter, and Values is (Count of) CustomerID. We rename the resulting worksheet **All cohorts -- num active**.
- We repeat this, creating three additional pivot tables where Values is (Sum of) NumTrans, Spend, and Profit, renaming the resulting worksheets **All cohorts -- tot trans**, **All cohorts -- tot spend**, and **All cohorts -- tot profit**.
- We copy the output from **All cohorts -- num active** (including row and column headings but not the “Grand Total” row and column) to cells A3:Q20 of a new worksheet, which we rename **All cohorts -- combined**, in a new workbook. (Don’t forget to save this new workbook, say `lens_4_analyses.xlsx`)
- Similarly, we copy the relevant output from **All cohorts -- tot trans** to cells A25:Q42 of this new worksheet, the relevant output of **All cohorts -- tot spend** to cells A47:Q64, and the relevant output of **All cohorts -- tot profit** to cells A69:Q86.

While we will consider two sets of cohort comparisons, let us create the elements of our multiplicative decomposition of cohort profit in a given period for all cohorts.

The first element is the percentage of the cohort that is active in that period.

- In order to compute this, we need to know the size of each cohort. This is given by the diagonal of the “number of active customers” table in cells A3:Q20. However, we do ignore the row associated with the “pre 2016” group of customers as we do not know the size of this cohort; we simply know that they were acquired some time before 2016.
- We can extract the diagonal using the `OFFSET` function in the following manner. We enter the label “cohort size” in cell S3,

=OFFSET(\$B\$5,ROW()-ROW(\$B\$5),ROW()-ROW(\$B\$5)) in cell S5, and copy the formula down to cell S20.

- We will compute the “% active” numbers in cells U3:AK19. We first copy the “y2016\_q1”, ..., “y2019\_q4” labels from cells B3:Q3 to cells U4:U19 and V3:AK3. We enter =IF(B5=0,"",B5/\$S5) in cell V4, format the cell as percentage, and copy the formula across and down to cell AK19.

The second element of our multiplicative decomposition is the average order frequency (AOF), which is simply total transactions divided by the number of active customers.

- First we copy the row and column headings associated with the number of transactions table across to cells U26:U42 and cells V25:AK25. Note that we now include the “pre 2016” cohort.
- We enter =IF(B26=0,"",B26/B4) in cell V26 and copy the formula across and down to cell AK42.

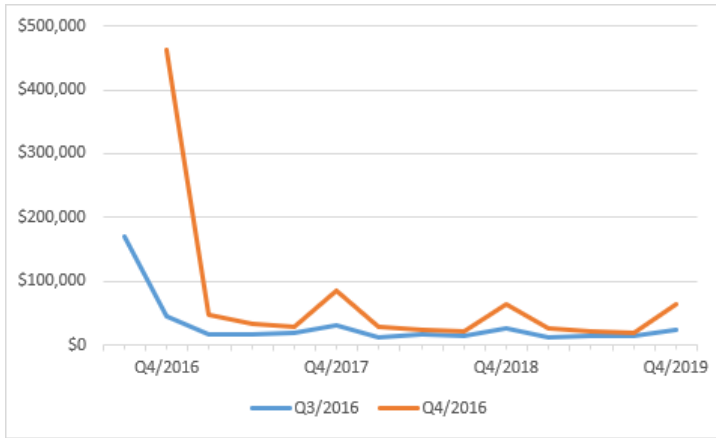
Copying U25:AK42 to U47:AK64 gives us the average order value (AOV) table. Similarly, copying U25:AK42 to U69:AK86 gives us the average margin table.

## 5.2 Comparing and Contrasting the Q3/2016 and Q4/2016 Cohorts

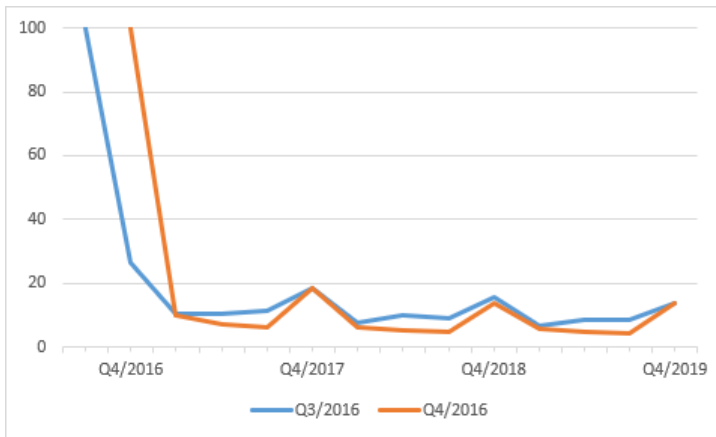
The starting point in comparing the performance of two cohorts is to plot their profit over time. We create Figure 5.1 by plotting the data in cells D73:Q73 and E74:Q74 on the same axes.

As we try to understand the difference in profits, we need to account for the fact that the Q4/2016 cohort is over twice the size of the Q3/2016 cohort (6,162 vs. 2,842 individuals). One possible solution would be to index the profit associated with each cohort in each period relative to the profit associated with their acquisition period (i.e., divide each element of D73:Q73 by D73 (and multiply by 100) and each element of E74:Q74 by E74). This results in Figure 5.2.

However, such an analysis gives us no real insight as to why the performance of one cohort is different from another. Controlling for the differences in cohort size, we use the multiplicative decomposition of profit developed above and examine the evolution of % cohort active, AOF, AOV and average margin.

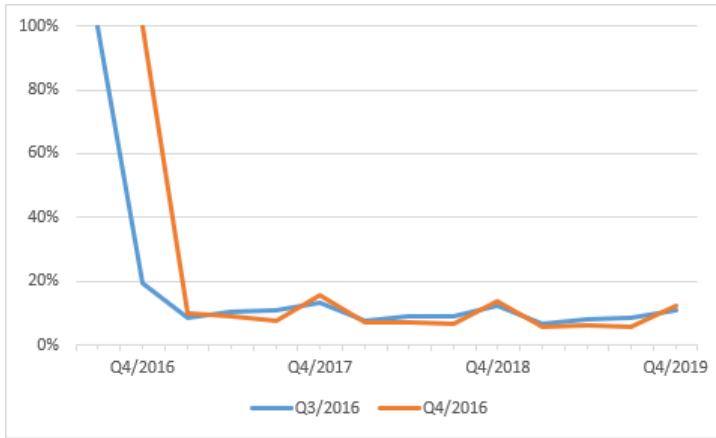


**Figure 5.1:** Quarterly profit associated with the Q3/2016 and Q4/2016 cohorts

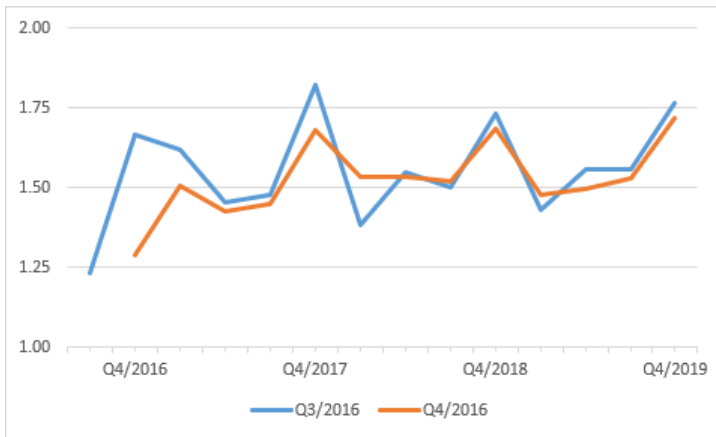


**Figure 5.2:** Quarterly profit associated with the Q3/2016 and Q4/2016 cohorts indexed against acquisition-quarter profits

- For % cohort active, we create Figure 5.3 by plotting the data in cells X6:AK7.
- For AOF, we create Figure 5.4 by plotting the data in cells X29:AK30.

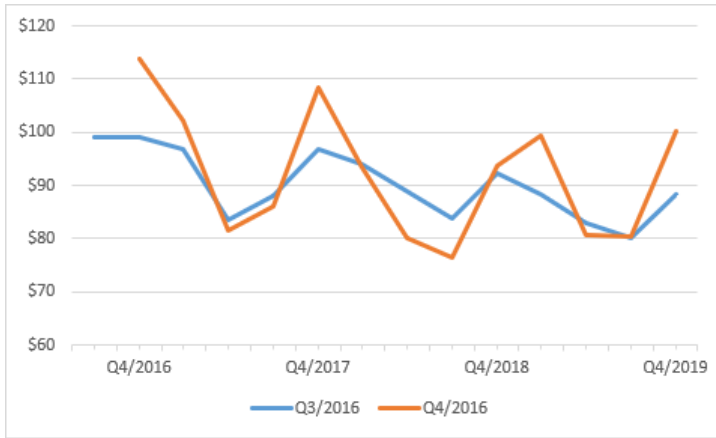


**Figure 5.3:** Percentage of the Q3/2016 and Q4/2016 cohorts active each quarter

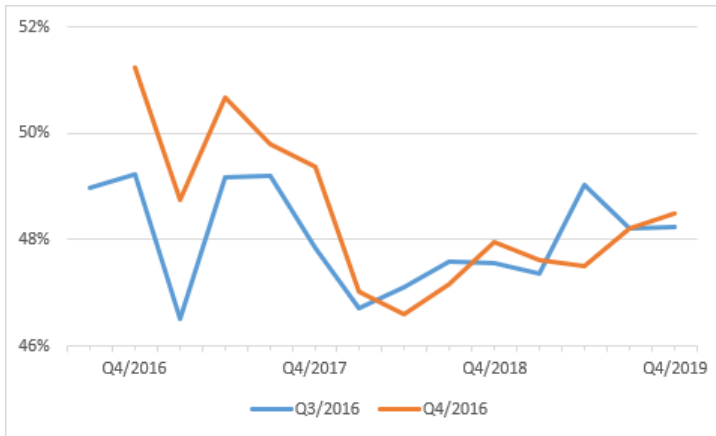


**Figure 5.4:** Evolution of AOF for the Q3/2016 and Q4/2016 cohorts

- For AOV, we create Figure 5.5 by plotting the data in cells X51:AK52.
- For average margin, we create Figure 5.6 by plotting the data in cells X73:AK74.



**Figure 5.5:** Evolution of AOV for the Q3/2016 and Q4/2016 cohorts



**Figure 5.6:** Evolution of average margin for the Q3/2016 and Q4/2016 cohorts

### 5.3 Comparing and Contrasting the Q4/2016 and Q4/2017 Cohorts

We leave the creation of the corresponding plots for the Q4/2016 and Q4/2017 cohorts (as presented in *TCBA*, Figures 6.8–6.13) as an exercise for the reader.

## Chapter 6

# Lens 5 Analyses

Lens 5 sees us undertaking a series of analyses designed to give us insight into the “health” of the firm’s customer base. In this chapter we show how to perform all the analyses presented in Chapter 7 of *TCBA*.

### 6.1 Creating the Working Datasets

Most of figures and tables we will create as part of our Lens 5 analyses using annual cohorts are based off the summaries presented in Tables 6.1–6.4. Table 6.1 reports the number of people in each annual cohort — in this case pre-2016, 2016, 2017, 2018, and 2019 — that were active in each year. Table 6.2 reports the total number of transactions made by the members of each annual cohort in each of 2016–2019. Tables 6.3 and 6.4 report the corresponding total spend and total profit numbers.

|           | 2016  | 2017  | 2018  | 2019  |
|-----------|-------|-------|-------|-------|
| pre_y2016 | 5756  | 4616  | 4242  | 3913  |
| y2016     | 14917 | 4092  | 3495  | 3064  |
| y2017     | 0     | 12726 | 3352  | 2714  |
| y2018     | 0     | 0     | 15165 | 3847  |
| y2019     | 0     | 0     | 0     | 18317 |
|           | 20673 | 21434 | 26254 | 31855 |

**Table 6.1:** Number of customers active each year by annual cohort

We will start with Table 6.1, which is the most difficult to create.

|           | 2016  | 2017  | 2018  | 2019  |
|-----------|-------|-------|-------|-------|
| pre_y2016 | 13535 | 11138 | 11019 | 10284 |
| y2016     | 23093 | 9830  | 8738  | 7596  |
| y2017     |       | 19068 | 8057  | 6504  |
| y2018     |       |       | 22974 | 9175  |
| y2019     |       |       |       | 27171 |
|           | 36628 | 40036 | 50788 | 60730 |

**Table 6.2:** Total annual transactions by annual cohort

|           | 2016    | 2017    | 2018    | 2019    |
|-----------|---------|---------|---------|---------|
| pre_y2016 | 1367752 | 1089809 | 1021231 | 971075  |
| y2016     | 2396133 | 925618  | 780723  | 681326  |
| y2017     |         | 1970194 | 728659  | 592696  |
| y2018     |         |         | 2284890 | 829307  |
| y2019     |         |         |         | 2762308 |
|           | 3763885 | 3985621 | 4815502 | 5836712 |

**Table 6.3:** Total annual spend by annual cohort

|           | 2016    | 2017    | 2018    | 2019    |
|-----------|---------|---------|---------|---------|
| pre_y2016 | 678387  | 536888  | 488756  | 470252  |
| y2016     | 1193524 | 451670  | 369951  | 328007  |
| y2017     |         | 964671  | 342182  | 284142  |
| y2018     |         |         | 1089454 | 387277  |
| y2019     |         |         |         | 1329226 |
|           | 1871911 | 1953229 | 2290343 | 2798904 |

**Table 6.4:** Total annual profit by annual cohort

- We open `cust_data_long.csv` in Excel and save it as a `xlsx` file (e.g., `creating_lens_5_data.xlsx`).
- Column C tells us the YearQuarter associated with each record. We want to create a variable that indicates the calendar year associated with each row of data. We enter “Year” in cell G1, `=LEFT(C2,5)` in cell G2, and copy the formula down to G130790.
- The next step is to create a table that indicates whether each customer was active (1) or not (0) each year.
  - We first insert a pivot table where Rows is CustomerID, Columns is Year, and Values is (Sum of) NumTrans. We rename the resulting worksheet **wip**.

- We will need a “clean” copy of this pivot-table output in our analyses. We enter the labels “CustomerID”, “2016”, “2017”, “2018”, and “2019” in cells H4:L4, =A5 in cell H5, and copy this formula across and down to cell L70045.
- The next step is to create a table that indicates whether each customer was active (1) or not (0) each year. We enter the labels “2016”, “2017”, “2018”, and “2019” in cells N4:Q4. We enter =1\*(B5>0) in cell N5 and copy this formula across and down to cell Q70045.
- We now have an indicator of whether each customer was active in each year. We want to sum up the column by annual cohort, which means we need to add a column that indicates which annual cohort each individual belongs to.
- In order to identify each customer’s annual cohort, we back back to **cust\_data\_long** and insert a pivot table where Rows is CustomerID, Columns is Cohort, and Values is (Sum of) NumTrans. We rename the resulting worksheet **Cohort mapping**. Looking at the structure of the pivot table, we see that each customer has only one column entry. We want to extract the heading of this column.
  - First, we enter “CustomerID” in cell U4, =A5 in cell U5, and copy this formula down to cell U70045. Next, we enter “Cohort” in cell V4.
  - We will extract the column heading using the INDEX and MATCH functions. For each customer, we identify which column contains a non-zero entry and extract the associated column heading from row 4. If you are using a recent version of Excel for Microsoft 365 or (at least) Excel 2021), enter =INDEX(B\$4:R\$4,MATCH(TRUE,(B5:R5<>0),0)) in cell V5 and copy this formula down to cell V70045.<sup>1</sup>
  - We identify the annual cohort associated with each customer’s quarterly cohort by entering “Cohort\_Year” in cell W4, =IF(V5="pre\_y2016",V5,LEFT(V5,5)) in cell W5, and copying this formula down to cell W70045.

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<sup>1</sup>If you are using an older version of Excel enter =INDEX(B\$4:R\$4,MATCH(TRUE,(B5:R5<>0),0)) in cell V5 and press Ctrl+Shift+Enter to convert this into an array formula. You can then copy the formula down to cell V70045.

- We copy and paste values the data in cells W4:W70045 to cells R4:R70045 in **wip**.
- We are finally in a position to count the number of (annual) cohort members that are active each calendar year. With the active cell somewhere in cells N4:R70045, we insert a pivot table where Rows is Cohort\_year and Values is (Sum of) 2016, 2017, 2018, and 2019. We rename this new worksheet **Annual cohorts -- num active**. We copy the pivot table output (including row and column headings and the “Grand Total” row but not the “Grand Total” column) to cells A3:E9 of a new worksheet, which we rename **Annual cohorts -- combined**, in a new workbook. (Don’t forget to save this new workbook, say `lens_5_analyses.xlsx`.) This gives us Table 6.1.
- While we are at it, let us copy cells H4:L70045 and R4:R70045 of **wip** to a new worksheet in this new workbook, and rename the worksheet **Customer annual activity**.

Creating the remaining three tables is a more straight-forward exercise.

- Going back to **cust\_data\_long** in `creating_lens_5_data.xlsx`, we start by creating a variable that indicates the annual cohort to which the customer associated with each row of data belongs. We enter “Cohort\_Year” in cell H1, `=IF(B2="pre_y2016",B2,LEFT(B2,5))` in cell H2, and copy the formula down the column to cell H130790.
- We create Table 6.2 by inserting a pivot table where Rows is Cohort\_Year, Columns is Year, and Values is (Sum of) NumTrans. We rename the new worksheet **Annual cohorts -- num trans** and copy the pivot table output (including row and column headings but not the “Grand Total” column) to cells A14:E20 of **Annual cohorts -- combined** (in `lens_5_analyses.xlsx`).
- We create Table 6.3 in a similar manner. Going back to **cust\_data\_long**, we insert a pivot table where Rows is Cohort\_Year, Columns is Year, and Values is (Sum of) Spend. We rename the new worksheet **Annual cohorts -- spend** and copy the pivot table output (including row and column headings and the “Grand Total” row but not the “Grand Total” column) to cells

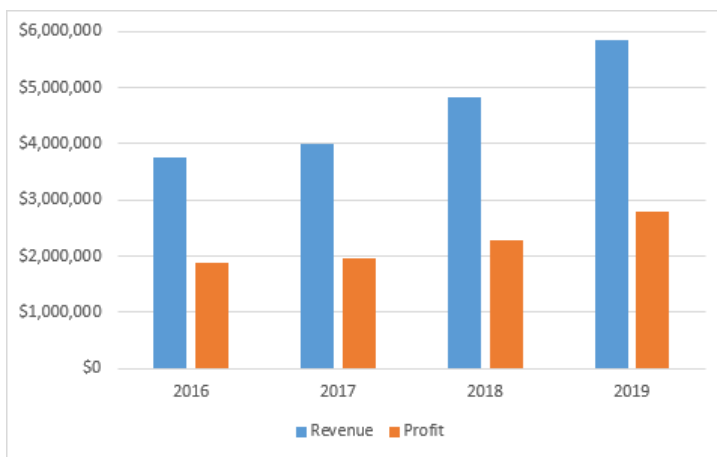
A25:E31 of **Annual cohorts -- combined**, rounding the numbers to zero decimal places.

- We repeat this to create Table 6.4, with Values being (Sum of) Profit. We rename the new worksheet **Annual cohorts -- profit** and copy the pivot table output (including row and column headings and the “Grand Total” row but not the “Grand Total” column) to cells A36:E42 of **Annual cohorts -- combined**, rounding the numbers to zero decimal places.

We can now close `creating_lens_5_data.xlsx` and focus on `lens_5_analyses.xlsx`. We make a copy of **Annual cohorts -- combined**, which Excel automatically renames **Annual cohorts -- combined (2)**, and will start working in this new worksheet.

## 6.2 Examining Annual Performance

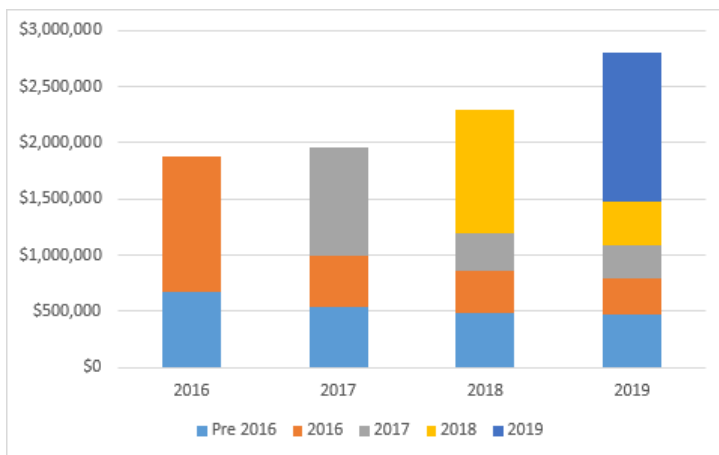
The simplest summary of firm performance is Figure 6.1, which plots annual revenue and profit. This is simply a bar chart created using total spend (as given in cells B31:E31) and total profit (as given in cells B42:E42).



**Figure 6.1:** Summary of annual performance

Figure 6.2 decomposes each year’s profit into that attributable to each (annual) acquisition cohort, and is a plot of Table 6.4. Within Excel, it is a 2-D Column / Stacked Column chart created using the

the data in cells A36:E41. Note that Excel automatically creates a plot where each bar represents a row of the table. We need each bar to represent a column of the table. To make this change, right-click on the plot, “Select Data” and click on “Switch Row/Column”.



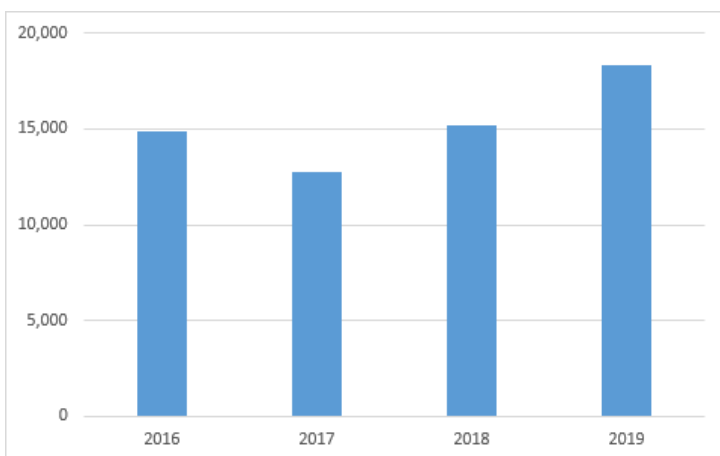
**Figure 6.2:** Annual profit by year of acquisition

Figure 6.3 reports the number of new customers acquired each year. The associated numbers come from the diagonal of Table 6.1. We can extract them into a single column in the following manner. We enter the label “cohort size” in cell G3, `=OFFSET($B$5,ROW()-ROW($B$5),ROW()-ROW($B$5))` in cell G5, and copy the formula down the column to cell G8.

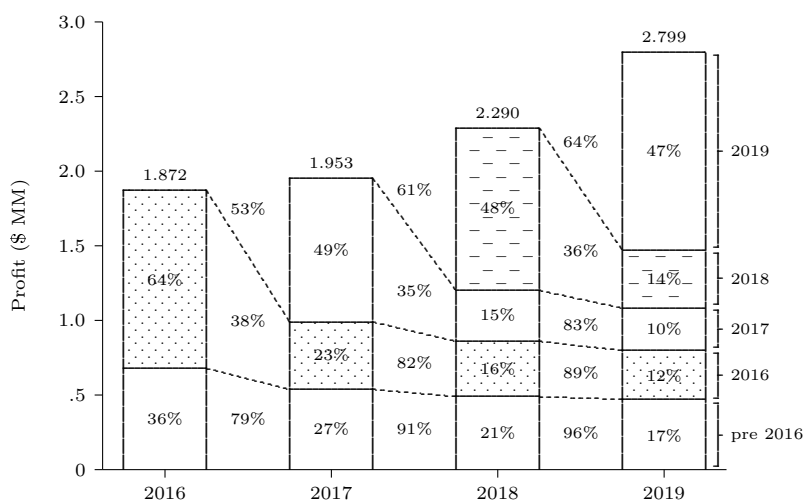
Figure 6.4 is a reworking of Figure 6.2 that presents the various percentage changes associated with the annual performance.

Let us consider the bar associated with 2016. With reference to Table 6.4, the proportion of that year’s profit attributed to customers acquired that year is  $\$1,193,524/\$1,871,911 = 0.64$ . This implies 36% of the profit in 2016 is attributable to customers acquired in previous years. The breakdown for the following three years is computed in the same manner.

Let us now consider the percentages between the bars associated with 2016 and 2017. Where does the 53% come from and what does it mean? The profit associated with those customers acquired in 2016 and in previous years is \$1,871,911. Looking at 2017, the total profit is \$1,953,229. Noting that the profit associated with customers acquired that year is \$964,671, this means the profit associated with customers



**Figure 6.3:** Number of customers acquired each year



**Figure 6.4:** Decomposition of annual profit by year of acquisition

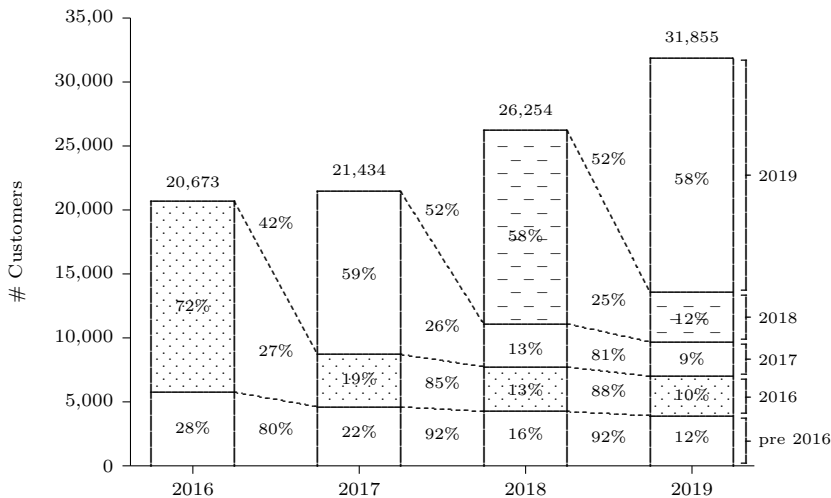
acquired in previous years is  $\$1,953,229 - \$964,671 = \$988,558$ , which is 53% of the profit associated with those same cohorts in the previous year ( $\$988,558 / \$1,871,911 = 0.53$ ).

What about 38% and 79%? The 2016 profit associated with customers acquired that year is  $\$1,193,524$ , while in 2017, the profit associated with this cohort of customers is  $\$451,670$ , which is 38% of the 2016 number. In a similar manner, the 2017 profit associated with

those customer acquired before 2016 is \$536,888, which is 79% of the profit associated with the same cohort in 2016 (\$678,387).

Having computed the various percentage breakdowns using the data in Table 6.4, Figure 6.4 is created using a drawing package.

We create Figure 6.5 in the same manner, using the data on the number of customers active year year given in Table 6.1.



**Figure 6.5:** Evolution of the number of active customers by year of acquisition

### 6.3 Exploring Annual Cohort Dynamics

Table 6.5 reports the development of the time to second purchases for each of the four annual cohorts. We create this table in the following manner.

|      | Year 1 | Year 2 | Year 3 | Year 4 |
|------|--------|--------|--------|--------|
| 2016 | 29%    | 43%    | 49%    | 52%    |
| 2017 | 27%    | 41%    | 47%    |        |
| 2018 | 27%    | 40%    |        |        |
| 2019 | 26%    |        |        |        |

**Table 6.5:** Cumulative percentage of each annual cohort making their second-ever purchase

- Let us return to **Customer annual activity**, which reports the number of transactions made by each customer each year.
- We want to create a table where reading across the row associated with each customer, a table entry of 0 indicates that a customer has yet to make a second purchase (which could be because they are associated with a cohort acquired after the year associated with a given column) and 1 indicates that a customer has made a second purchase (either in the current year or in a previous year).
- Having entered the column labels “y2016”, . . . , “y2019” in cells G1:J19, we accomplish this by entering `=IF($F2="pre_y2016", 0, IF($F2>G$1, 0, IF($F2=G$1, 1*(B2>1), MAX(F2, B2>0))))` in cell G2 and copying the formula across and down to cell J70042. The first IF sees us returning 0 for those customers acquired before 2016. The second IF means we return 0 if the cohort associated with the current row was born after the year associated with the current column. The third IF sees us checking if the cohort associated with the current row was born in the year associated with the current column. If the answer is yes, it flags a repeat purchase if the number of transactions in that year is greater than 1. Otherwise, it returns 1 if the number of transactions in this year is greater than 0 or if a second-ever purchase had been made in a previous year.
- It does not matter that we have computed these numbers for the “pre\_2016” cohort; we will address this below.
- We insert a pivot table where Rows is Cohort\_Year, and Values is (Sum of) y2016, y2017, y2018, and y2019. To remove the row associated with the “pre\_2016” cohort, we click on the drop-down filter arrow for Row Labels (cell A3) and deselect “pre\_2016”. We rename this new worksheet **Second-ever purchase**.
- Looking across the row associated with the 2016 cohort, we see that 4276 members of that cohort made their second-ever purchase in 2016. By the end of 2017, 6418 members of that cohort had made a second purchase, and so on. We want to present these numbers as a percentage of cohort members. We copy the

cohort size numbers from cells G5:G8 of **Annual cohorts -- combined** to cells G4:G7 of the current worksheet.

- We enter the year headings “2016”, . . . , “2019” in cells J3:M3. Next, enter the formula `=B4/$G4` in cell J4, format the cell as percentages, and copy the formula across and down to cell M7.
- We arrive at Table 6.5 by left-aligning the non-zero numbers in each row.

Table 6.6 reports the annual repeat-buying rate for each cohort and the whole customer base. We create this table in the following manner.

|          | 2016/17 | 2017/18 | 2018/19 |
|----------|---------|---------|---------|
| Pre 2016 | 51%     | 56%     | 57%     |
| 2016     | 27%     | 52%     | 55%     |
| 2017     |         | 26%     | 50%     |
| 2018     |         |         | 25%     |
| Overall  | 34%     | 38%     | 38%     |

**Table 6.6:** Annual repeat-buying rates

- Recall that the annual repeat-buying rate for a given cohort is the number of cohort members that made at least one purchase in both years  $x$  and  $x + 1$ , divided by the number of cohort members that made at least one purchase in year  $x$ . The first thing we need to do is compute the number of cohort members that made at least one purchase in each of the adjacent pairs of years. The **Customer annual activity** worksheet gives us a table that reports the number of transactions made by each customer each year. We enter the labels “16/17”, “17/18” and “18/19” in cells K1:M1. Then we enter `=(B2>0)*(C2>0)` in cell K2 and copy the formula across and down to cell M70042.
- We insert a pivot table where Rows is Cohort\_Year, and Values is (Sum of) 16/17, 17/18, and 18/19. We rename this worksheet **RBR**.
- We see that 2954 members of the pre 2016 cohort made at least one purchase in both 2016 and 2017, 2588 made at least one purchase in both 2017 and 2018, and so on. In order to convert

these numbers into repeat-buying rates, we need to divide these numbers by the number of members of that cohort that made at least one purchase in 2016, at least one purchase in 2017, and so on. We already have this information (Table 6.1). We copy cells A3:E9 from **Annual cohorts - - combined** to cells F3:J9 in **RBR**.

- We are now in a position to create Table 6.6. We enter the labels “2016/17”, “2017/18”, and “2018/19” in cells H11:J11, and the labels “Pre 2016”, “2016”, “2017”, “2018”, and “Overall” in cells G12:G16.<sup>2</sup> In order to compute the cohort-specific repeat-buying rates, we enter `=IF(G4>0,B4/G4,"")` in cell H12, format the cell as percentage, and copy the formula across and down to cell J15. (We wrap the calculation within an IF statement to avoid seeing #DIV/0! cells in the table.) We compute the overall repeat-buying rate (i.e., for all the customers, not broken down by cohort) by entering `=B9/G9` in cell H16, formatting the cell as percentage, and copying the formula across to cell J16.

## 6.4 Digging Deeper into Cohort Performance

In order to make more sense of the cohort-level decomposition of annual profit as presented in Figures 6.2 and 6.4, we can make use of multiplicative decompositions along the lines of those developed in our Lens 3 and 4 analyses.

Let us first consider Figures 6.6 and 6.7, which plot the percentage of each annual cohort that is active each year and the average annual profit per active annual cohort member, respectively.

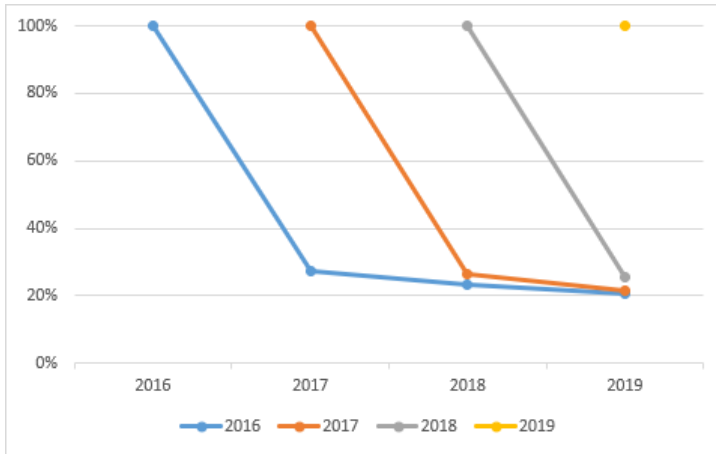
Figure 6.6 is a plot of the data presented in Table 6.7, which we create from Table 6.1 in the following manner.

|      | 2016 | 2017 | 2018 | 2019 |
|------|------|------|------|------|
| 2016 | 100% | 27%  | 23%  | 21%  |
| 2017 | #N/A | 100% | 26%  | 21%  |
| 2018 | #N/A | #N/A | 100% | 25%  |
| 2019 | #N/A | #N/A | #N/A | 100% |

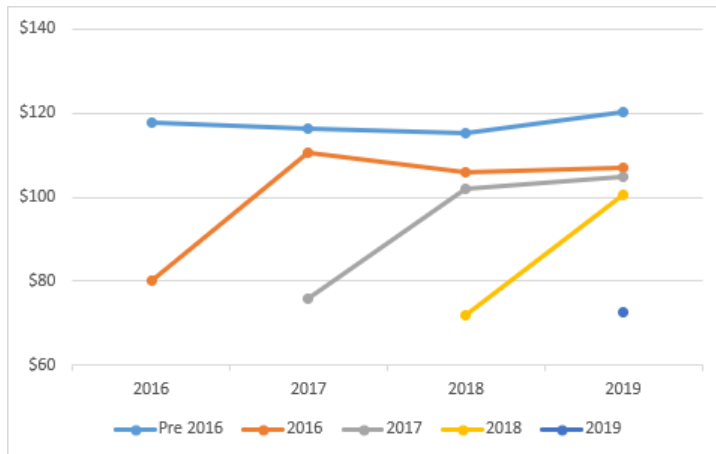
**Table 6.7:** Annual percent active by annual cohort

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<sup>2</sup>Why don’t we have a 2019 row?



**Figure 6.6:** Annual percent active by annual cohort



**Figure 6.7:** Average annual profit per active cohort member by annual cohort

- We make a copy of **Annual cohorts – combined**, which Excel automatically renames **Annual cohorts – combined (3)**, and will start working in this new worksheet.
- We extract the cohort sizes in cells G5:G8 (as computed above).
- As we do not know how many customers were acquired before 2016, we can only compute the percentage of the cohort that is active for those customers acquired in 2016–2019.

- We will compute the % active numbers in cells I3:M7. We first copy the “2016”, ..., “2019” labels to cells J3:M3 and I4:I7. We enter `=IF(B5>0,B5/$G5,NA())` in cell J4, format the cell as percentage, and copy the formula across and down to cell M7. Why do we use `NA()` and not `""` as before? If we use `""`, any resulting “blank” cell will be treated as a zero and plotted accordingly when we create the figure. On the other hand, the `#N/A` that follows from using `NA()` will be ignored by Excel when creating the figure.
- We create Figure 6.6 by selecting cells I3:M7 and inserting a 2-D Line / Line with Markers chart.

Figure 6.7 is a plot of the data presented in Table 6.8, which we create in the following manner.

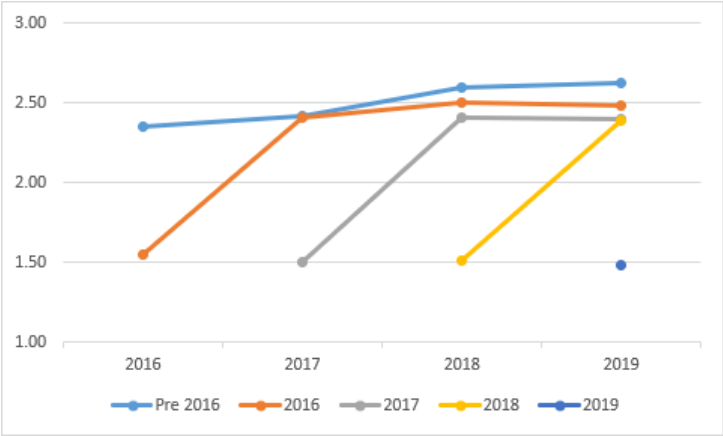
|          | 2016  | 2017  | 2018  | 2019  |
|----------|-------|-------|-------|-------|
| Pre 2016 | \$118 | \$116 | \$115 | \$120 |
| 2016     | \$80  | \$110 | \$106 | \$107 |
| 2017     | #N/A  | \$76  | \$102 | \$105 |
| 2018     | #N/A  | #N/A  | \$72  | \$101 |
| 2019     | #N/A  | #N/A  | #N/A  | \$73  |
| Total    | \$91  | \$91  | \$87  | \$88  |

**Table 6.8:** Average annual profit per active cohort member by annual cohort

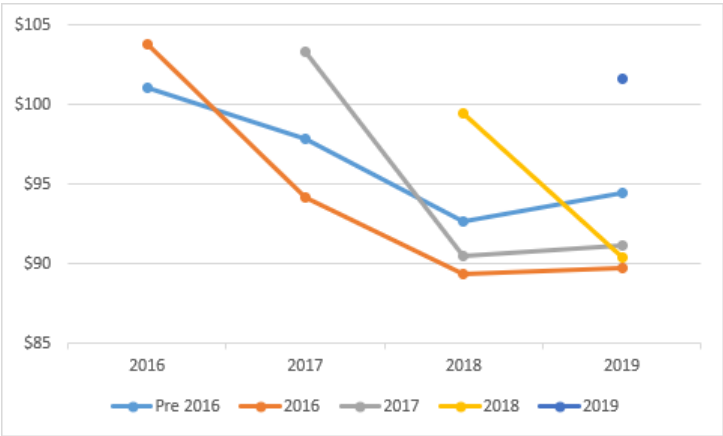
- We add the labels “2016”, ..., “2019” to cells J14:M14, and “Pre 2016”, “2016”, ..., “2019”, and “Total” in cells I15:I20.
- For any given year, the average annual profit per active cohort member is simply the annual profit associated with that cohort in that year (from Table 6.4) divided by the number of cohort members that were active that year (from Table 6.1). We enter `=IF(B4>0,B37/B4,NA())` in cell J15, format the cell as currency, and copy the formula across and down to cell M20.
- We create Figure 6.7 by selecting cells I14:M19 and inserting a 2-D Line / Line with Markers chart. (Note that we exclude the Total row (cells J20:M20).) Excel automatically creates a plot where each line represents a column of the table. However, we want each line to represent a row of the table. To make

this change, right-click on the plot, “Select Data” and click on “Switch Row/Column”.

We can decompose average annual profit per active cohort member into annual AOF  $\times$  annual AOV  $\times$  annual average margin, which we plot in Figures 6.8–6.10.

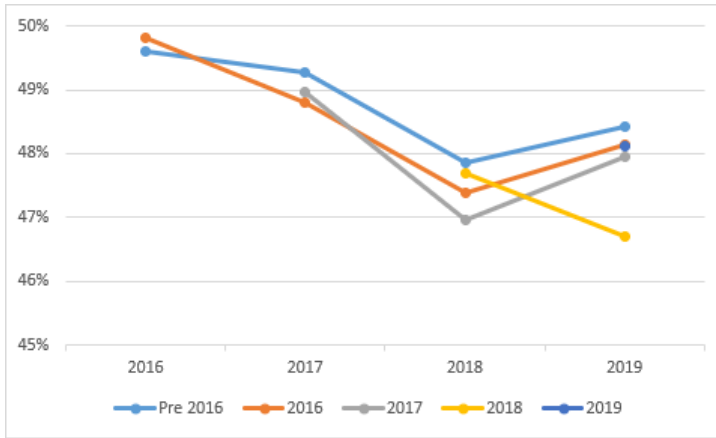


**Figure 6.8:** Annual AOF by annual cohort



**Figure 6.9:** Annual AOV by annual cohort

By now, you should be able to create the underlying data tables without any guidance, as the steps mirror those used above.



**Figure 6.10:** Annual average margin by annual cohort

- For AOF, we enter the relevant row and columns labels in cells I26:I31 and J25:M25, respectively. Next, we enter `=IF(B4>0,B15/B4,NA())` in cell J26 and copy the formula across and down to cell M31.
- For AOV, we enter the relevant row and columns labels in cells I37:I42 and J36:M36, respectively. Next, we enter `=IF(B15>0,B26/B15,NA())` in cell J37, format the cell as currency, and copy the formula across and down to cell M42.
- For average margin, we enter the relevant row and columns labels in cells I48:I53 and J47:M47, respectively. Next, we enter `=IF(B26>0,B37/B26,NA())` in cell J48, format the cell as percentage, and copy the formula across and down to cell M53.

Table 6.9 is simply the Total row from each of the three tables we have just created.

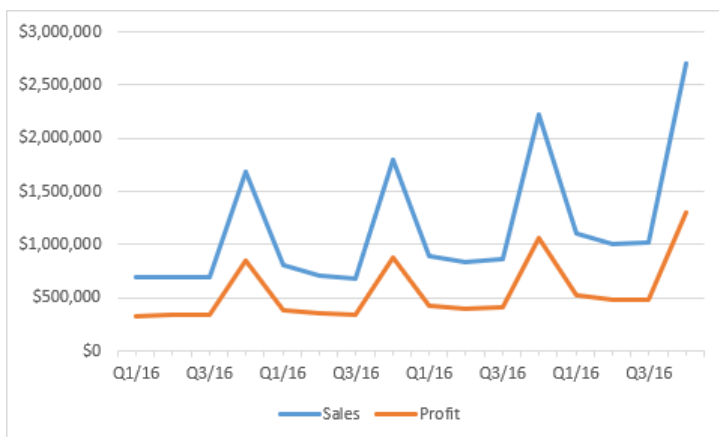
|            | 2016     | 2017    | 2018    | 2019    |
|------------|----------|---------|---------|---------|
| AOF        | 1.77     | 1.87    | 1.93    | 1.91    |
| AOV        | \$102.76 | \$99.55 | \$94.82 | \$96.11 |
| Avg margin | 50%      | 49%     | 48%     | 48%     |

**Table 6.9:** Overall AOF, AOV, and average margin by year

## 6.5 Quarterly Analysis

We can perform similar analyses at the quarter level using quarterly cohorts. Three basic figures are presented in *TCBA*, which we can create in the following manner.

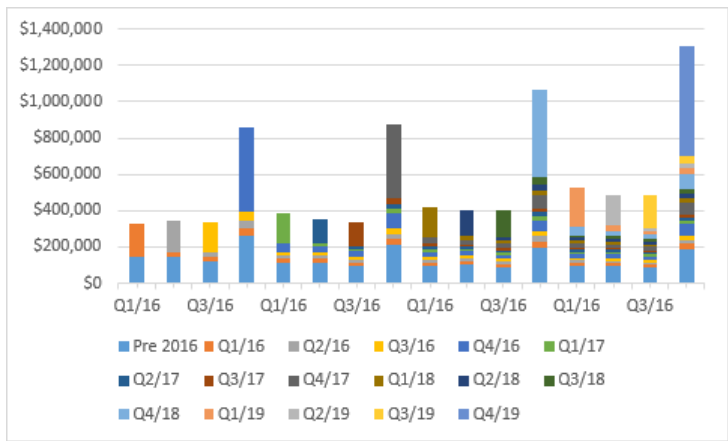
- We first copy the **All cohorts -- combined** worksheet from `lens_4_analyses.xlsx` into `lens_5_analyses.xlsx`
- Figure 6.11 reports Madrigal's sales and profit on a quarterly basis. The sales and profit tables in **All cohorts -- combined** do not include column totals. We create these in a new worksheet, which we call **Quarterly plots**. We first copy across the quarter labels (e.g., cells B47:Q47) from **All cohorts -- combined** to B1:Q1 of **Quarterly plots**. We compute total sales for each quarter by entering `=SUM('All cohorts -- combined'!B48:B64)` in cell B2 and copying the formula across to cell Q2. Entering `=SUM('All cohorts -- combined'!B70:B86)` in cell B3 and copying across to Q3 gives us quarterly profit. Figure 6.11 is simply a plot of these two rows of data.



**Figure 6.11:** Summary of quarterly performance

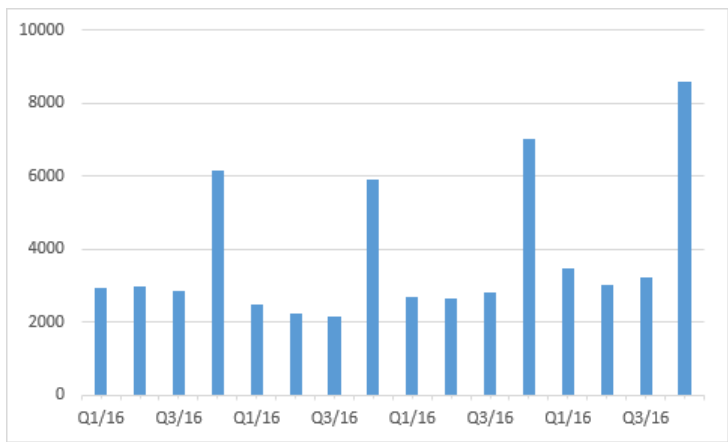
- Figure 6.12 decomposes each quarter's profit into the amount associated with the customers acquired that quarter versus previous cohorts. We copy the profit data in **All cohorts -- combined** (cells A69:Q86) into **Quarterly plots** (cells A8:Q25) and create a stacked 2-D column bar chart. Note that Excel

automatically creates a plot where each bar represents a row of the table. We need each bar to represent a column of the table. To make this change, right-click on the plot, “Select Data” and click on “Switch Row/Column”. (We have changed the row labels in cells A9:A25 so as to create a tidier plot.)



**Figure 6.12:** Decomposing quarterly profit by (quarterly) cohort

- Figure 6.13 is a plot of the number of customers acquired each cohort (i.e., the cohort size), which can be found in cells S5:S20 of **All cohorts -- combined**.



**Figure 6.13:** Number of customer acquired each quarter

## Appendix A

# Creating the Working Datasets from the “Wide” CSV Files

We have created the various working datasets required for the Lenses 1–5 analyses assuming customer  $\times$  quarter transaction summaries are provided as a “long” CSV file. In this appendix, we show how to create them when the data are provided as “wide” CSV files.

### A.1 Lens 1

We create the working dataset (as illustrated in Figure 2.1) from the “wide” datasets in the following manner.

- We open `cust_by_qtr_trans.csv` in Excel and save it as an `xlsx` file (e.g., `creating_lens_1_data_alt.xlsx`).
- We open `cust_by_qtr_spend.csv` and copy the worksheet **cust\_by\_qtr\_spend** into `creating_lens_1_data_alt.xlsx`.
- We open `cust_by_qtr_profit.csv` and copy the worksheet **cust\_by\_qtr\_profit** into `creating_lens_1_data_alt.xlsx`.
- We copy the CustomerID numbers from **cust\_by\_qtr\_trans** to column A of a new worksheet.
- The next step is to compute each customer’s total number of transactions in 2019. We enter “NumTrans” in cell B1, `=SUM(`

`cust_by_qtr_trans!02:R2)` in cell B2, and copy the formula down to cell B70042.

- We compute each customer’s total spend in 2019 by entering “Spend” in cell C1, `=SUM(cust_by_qtr_spend!02:R2)` in cell C2, and copying the formula down to cell C70042.
- We compute each customer’s 2019 profit by entering “Profit” in cell D1, `=SUM(cust_by_qtr_profit!02:R2)` in cell D2, and copying the formula down to cell D70042.
- We are only interested in those customers that made at least one transaction in 2019. We turn on filtering (Ctrl+Shift+L), click on the drop-down filter arrow for NumTrans (cell B1), and deselect 0. We copy columns A:D of the filtered table into a new workbook, renaming the worksheet **Data**. This table should be exactly the same as that reported in Figure 2.1.

Yes, it was easier to create this working dataset using the “long” version of the main dataset.

Note: We have assumed that the three spreadsheets contain the same customers IDs, sorted in the same order. In practice, we should never make such assumption. Check that it is true.

## A.2 Lens 2

We create the working dataset (as illustrated in Figure 3.1) from the “wide” datasets in the following manner.

- We open `cust_by_qtr_trans.csv` in Excel and save it as an `xlsx` file (e.g., `creating_lens_2_data_alt.xlsx`).
- We open `cust_by_qtr_spend.csv` and copy the worksheet **cust\_by\_qtr\_spend** into `creating_lens_2_data_alt.xlsx`.
- We open `cust_by_qtr_profit.csv` and copy the worksheet **cust\_by\_qtr\_profit** into `creating_lens_2_data_alt.xlsx`.
- We copy the CustomerID numbers from **cust\_by\_qtr\_trans** to column A of a new worksheet.

- The first step is to compute each customer’s total number of transactions, total spend, and total profit in 2018. We enter “trans.2018” in cell B1, `=SUM(cust_by_qtr_trans!K2:N2)` in cell B2, and copy the formula down to cell B70042. Next we enter “spend.2018” in cell C1, `=SUM(cust_by_qtr_spend!K2:N2)` in cell C2, and copy the formula down to cell C70042. Finally, we enter “profit.2018” in cell D1, `=SUM(cust_by_qtr_profit!K2:N2)` in cell D2, and copy the formula down to cell D70042.
- We repeat this for 2019. We enter “trans.2019” in cell E1, `=SUM(cust_by_qtr_trans!O2:R2)` in cell E2, and copy the formula down to cell E70042. Next we enter “spend.2019” in cell F1, `=SUM(cust_by_qtr_spend!O2:R2)` in cell F2, and copy the formula down to cell F70042. Finally, we enter “profit.2019” in cell G1, `=SUM(cust_by_qtr_profit!O2:R2)` in cell G2, and copy the formula down to cell G70042.
- We are only interested in those customers that made at least one transaction in 2018 or 2019, which means we want to remove the rows associated with those customers that made zero transactions across both years. We create a dummy variable that takes on the value of 1 if the customer made at least one transaction in either year and a value of 0 if they purchased in neither year. We enter “Use?” in cell H1, `=1*(B2+E2>0)` in cell H2, and copy the formula down to H70042. Next we turn on filtering (Ctrl+Shift+L), click on the drop-down filter arrow for Use? (cell H1) and deselect 0. We copy columns A:G of the filtered table into a new workbook, renaming the worksheet **Data**. This table should be exactly the same as that reported in Figure 3.1.

### A.3 Lens 3

Our Lens 3 analyses make use of several working datasets. The first step in creating them is to extract the records associated with the Q1 2016 cohort.

- We open `cust_by_qtr_trans.csv` in Excel, turn on filtering (Ctrl+Shift+L), click on the drop-down filter arrow for Cohort, and deselect all the options except “y2016\_q1”. (The easiest way to do this is to untick (“Select All”) and then select “y2016\_q1”.)

- We copy columns A:R of the filtered table into a worksheet in a new workbook, which we rename **Trans**. There should be 2945 rows in this worksheet. (Don't forget to save this new workbook, say `creating_lens_3_data_alt.xlsx`)
- We repeat this for `cust_by_qtr_spend.csv`, copying the filtered data into a new worksheet **Spend** in the workbook `creating_lens_3_data_alt.xlsx`
- We repeat this for `cust_by_qtr_profit.csv`, copying the filtered data into a new worksheet **Profit** in the workbook `creating_lens_3_data_alt.xlsx`

### A.3.1 Part 1

We create the quarterly summary of cohort-level behavior in the following manner.

- We add a new worksheet in `creating_lens_3_data_alt.xlsx`, renaming it **Quarterly summary**.
- We copy the quarter labels (cells C1:R1) of **Trans** to cells B1:Q1 of **Quarterly summary**.
- We enter the row labels “Quarter”, “NumActive”, “TotTrans”, and “TotSpend” in cells A1:A4.
- We determine the number of cohort members active each quarter by entering `=COUNTIF(Trans!C2:C2945,">0")` in cell B2 and copying the formula across to Q2.
- We determine the total number of transactions made by the cohort each quarter by entering `=SUM(Trans!C2:C2945)` in cell B3 and copying the formula across to Q3.
- We determine the total spend by the cohort each quarter by entering `=SUM(Spend!C2:C2945)` in cell B4 and copying the formula across to Q4.

Figure 4.1 is the transpose of this table.

### A.3.2 Part 2

There is no need for us to create the **Data (a)** summary; it is simply the worksheet **Trans**.

### A.3.3 Part 3

We create the four-year summary dataset (as illustrated in Figure 4.12) in the following manner.

- We insert a new worksheet, which we rename **Data (b)**, and add the columns names “CustomerID”, “SumNumTrans”, “SumSpend”, and “SumProfit” in cells A1:D4
- We copy the CustomerID numbers from **Trans** to column A.
- We compute each cohort member’s total number of transactions across the four-year period by entering =SUM(Trans!C2:R2) in cell B2 and copying the formula down to B2945.
- We compute each cohort member’s total spend across the four-year period by entering=SUM(Spend!C2:R2) in cell C2 and copying the formula down to C2945.
- We compute each cohort member’s total profit across the four-year period by entering =SUM(Profit!C2:R2) in cell D2 and copying the formula down to D2945.

## A.4 Lens 4

We create the worksheet that contains the cohort-by-quarter summaries of the number of active customers, the total number of transactions, total spend, and total profit in the following manner. (Note: This assumes you are using Excel for Microsoft 365 or (at least) Excel 2021.)

- We open `cust_by_qtr_trans.csv` in Excel and save it as an `xlsx` file (e.g., `creating_lens_4_data.alt.xlsx`).
- We open `cust_by_qtr_spend.csv` and copy the worksheet **cust\_by\_qtr\_spend** into `creating_lens_4_data.alt.xlsx`
- We open `cust_by_qtr_profit.csv` and copy the worksheet **cust\_by\_qtr\_profit** into `creating_lens_4_data.alt.xlsx`
- We add a new worksheet, which we rename **All cohorts -- combined**.

- We start by creating the row and column headings associated with the summary tables. The row labels are the levels of Cohort. We extract them by entering `=UNIQUE(cust_by_qtr_trans!B2:B70042)` in cell A4 and cells A4:A20 will be automatically populated with the unique cohort labels.
- An aside: It is important that the cohort labels are sorted in chronological order, from oldest to newest. When creating this dataset, labels were chosen so that this would happen automatically. For example, we use `y2016.q1` as opposed to `q1.y2016`.
- In order to create the (calendar time) column headings, we copy and paste the values of A5:A20 to B3:Q3.
- As we will be creating four summary tables, let us make three copies of these row and column headings. We copy A3:Q20 (and paste values) to A25:Q42, A47:Q64, and A69:Q86.
- Let us first consider how we count the number of active customers each quarter by cohort. Thinking back to **cust\_by\_qtr\_trans**, we want to consider just those customers associated with a particular cohort and then count the number that have at least one transaction in each quarter. We perform this calculation by entering `=SUM((cust_by_qtr_trans!B$2:$B$70042=$A4)*(cust_by_qtr_trans!C$2:C$70042>0))` in cell B4 and copying the formula across and down to cell Q20. (Do pay careful attention to the fixed versus relative row and column references.)
- A minor tweak to this formula gives us the total number of transactions made by each cohort each quarter. We delete `> 0` from the end of the formula, enter `=SUM((cust_by_qtr_trans!B$2:$B$70042=$A4)*(cust_by_qtr_trans!C$2:C$70042))` in cell B26 and copy the formula across and down to cell Q42.
- We use the same logic to compute each cohort's quarterly total spend, referencing **cust\_by\_qtr\_spend** instead of **cust\_by\_qtr\_trans**. We enter `=SUM((cust_by_qtr_spend!B$2:$B$70042=$A4)*(cust_by_qtr_spend!C$2:C$70042))` in cell B48 and then copy the formula across and down to cell Q64.
- For total profit, we enter `=SUM((cust_by_qtr_profit!B$2:$B$70042=$A4)*(cust_by_qtr_profit!C$2:C$70042))` in cell B70 and copy the formula across and down to cell Q86.

## A.5 Lens 5

We create the required annual cohort summaries in the following manner.

- We open `cust_by_qtr_trans.csv` in Excel and save it as an `xlsx` file (e.g., `creating_lens_5_data_alt.xlsx`).
- We open `cust_by_qtr_spend.csv` and copy the worksheet `cust_by_qtr_spend` into `creating_lens_5_data_alt.xlsx`.
- We repeat this for `cust_by_qtr_profit.csv`.
- Going to the worksheet `cust_by_qtr_trans` in `creating_lens_5_data_alt.xlsx`, we enter the labels “Cohort\_Year”, “2016”, “2017”, “2018”, and “2019” in cells T1:X1.
- We identify each customer’s annual cohort by entering `=IF(B2="pre_y2016",B2,LEFT(B2,5))` in cell T2 and copying the formula down to the column to cell T70042.
- We compute the first customer’s total number of transactions in 2016 by entering `=SUM(C2:F2)` in cell U3. The associated totals for 2017–2019 are computed by entering `=SUM(G2:J2)`, `=SUM(K2:N2)`, and `=SUM(O2:R2)` in cells V3:X3 respectively. We then copy this set of four cells down to row 70042.
- We compute these same quantities for spend and profit by copying cells T1:X70042 into the same block in `cust_by_qtr_spend` and `cust_by_qtr_profit`.
- Going back to `cust_by_qtr_trans`, we want to create a table that indicates whether each customer was active (1) or not (0) each year. We enter the labels “active 2016”, “active 2017”, “active 2018”, and “active 2019” in cells Y1:AB1. Next, we enter `=1*(U2>0)` in cell Y2 and copy this formula across and down to cell AB70042.
- Staying in `cust_by_qtr_trans` and with the active cell somewhere in cells T1:AB70042, we create the pivot table associated with Table 6.1 by inserting a pivot table where Rows is Cohort\_year and Values is (Sum of) active 2016, active 2017, active 2018, and active 2019. We rename this new worksheet **Annual cohorts -- num active**.

- The worksheet **Customer annual activity** (as used in Chapter 6) is created from columns A, U:X, and T (in that order) of **cust\_by\_qtr\_trans**.
- Going back to **cust\_by\_qtr\_trans** and with the active cell somewhere in cells T1:AB70042, we create the pivot table associated with Table 6.2 by inserting a pivot table where Rows is Cohort\_year and Values is (Sum of) 2016, 2017, 2018, and 2019. We rename this new worksheet **Annual cohorts -- num trans**.
- We create the pivot table associated with Table 6.3 by going to **cust\_by\_qtr\_spend** and, with the active cell somewhere in cells T1:X70042, inserting a pivot table where Rows is Cohort\_year and Values is (Sum of) 2016, 2017, 2018, and 2019. We rename this new worksheet **Annual cohorts -- spend**.
- We create the pivot table associated with Table 6.4 by applying the same process to **cust\_by\_qtr\_profit** and renaming the new worksheet **Annual cohorts -- profit**.

## Appendix B

# Creating an Area-proportional Venn Diagram

We note from Chapter 3 that Madrigal had 26,254 active customers in 2018 and 31,855 active customers in 2019, with 9,871 individuals active in both years. We want to visualize this using a Venn diagram where the area of each circle is proportional to the number of customers associated with that year and the area of the overlap is proportional to the number of customers active in both years. In order to create this, we need to compute the radius of each circle ( $R$  and  $r$ ), and the distance between the centers of the two circles ( $d$ ).

Determining the radius of each circle is simple. Let  $R$  be the radius of the circle associated with 2018 and  $r$  the radius of the circle associated with 2019. Considering the ratio of the number of active customers in 2019 to the number of customers active in 2018, we have

$$\frac{31,855}{26,254} = \frac{\pi r^2}{\pi R^2}$$

Without loss of generality, let the radius of the first circle equal one, which implies

$$\begin{aligned} r &= \sqrt{\frac{31,855}{26,254}} \\ &= 1.1015 \end{aligned}$$

Therefore, the area of the circle associated with 2018 is  $\pi$ , the area of the circle associated with 2019 is  $\pi(1.1015)^2$ , and area of the overlap between the two circles is  $\pi(9,871/26,254)$ .

How do we determine  $d$ ? We know that the area of overlap  $A$  for two circles of radii  $R$  and  $r$  centered at  $(0, 0)$  and  $(d, 0)$ , respectively, is given by<sup>1</sup>

$$A = r^2 \cos^{-1} \left( \frac{d^2 + r^2 - R^2}{2dr} \right) + R^2 \cos^{-1} \left( \frac{d^2 - r^2 + R^2}{2dR} \right) - \frac{1}{2} \sqrt{(d + r - R)(d - r + R)(-d + r + R)(d + r + R)}. \quad (\text{B1})$$

We can work backwards: Given  $R$ ,  $r$ , and  $A$ , we will solve for  $d$  numerically. With reference to Figure B1, we do this in the following manner.

- Given the customer numbers in cells B2:B4, we set the radius of the 2018 circle to 1 and determine the radius of the 2019 circle in cell B7.
- Having computed the area of the associated circles in cells C2:C3, we compute the “standardized” volume of the area of overlap in C4.
- Entering a starting value of  $d = 1$  in cell B8, we enter the formula given in equation (B1) in cell B9.
- We need to find the value of  $d$  that equates the computed value of  $A$  (cell B9) with the actual value of  $A$  (cell C4). We do this by computing the squared difference between these two cells (cell B13) and using Solver to find the value of  $d$  that minimizes this (to a value of zero) — see Figure B2. For this example, the solution is 1.1469.

This means we create Figure 3.3 by drawing a circle with radius 1 centered at  $(1, 1.1015)$  and a second circle with radius 1.1015 centered at  $(2.1469, 1.1015)$ . More generally, we will draw a circle of radius 1 centered at  $(1, \max(1, r))$  and a second circle with radius  $r$  centered at  $(1 + d, \max(1, r))$ .

---

<sup>1</sup><https://mathworld.wolfram.com/Circle-CircleIntersection.html>

|    | A        | B       | C                                                         | D              | E | F | G | H |
|----|----------|---------|-----------------------------------------------------------|----------------|---|---|---|---|
| 1  |          | # cust. | area                                                      |                |   |   |   |   |
| 2  | 2018     | 26254   | 3.1416                                                    | <-- =PI()      |   |   |   |   |
| 3  | 2019     | 31855   | 3.8118                                                    | <-- =PI()*B7^2 |   |   |   |   |
| 4  | both     | 9871    | 1.1812                                                    | <-- =B4/B2*C2  |   |   |   |   |
| 5  |          |         |                                                           |                |   |   |   |   |
| 6  | R        | 1       |                                                           |                |   |   |   |   |
| 7  | r        | 1.1015  | <-- =SQRT(B3/B2)                                          |                |   |   |   |   |
| 8  | d        | 1.0000  |                                                           |                |   |   |   |   |
| 9  | A        | 1.4453  | <-- =B7^2*ACOS((B8^2+B7^2-B6^2)/(2*B8*B7))+B6^2*          |                |   |   |   |   |
| 10 |          |         | ACOS((B8^2-B7^2+B6^2)/(2*B8*B6))-0.5*SQRT((B8+B7-B6)*(B8- |                |   |   |   |   |
| 11 |          |         | B7+B6)*(-B8+B7+B6)*(B8+B7+B6))                            |                |   |   |   |   |
| 12 |          |         |                                                           |                |   |   |   |   |
| 13 | sq. err. | 0.0698  | <-- =(C4-B9)^2                                            |                |   |   |   |   |

**Figure B1:** Excel sheet for determining  $R$ ,  $r$ , and  $d$

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Options

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Help  Close

**Figure B2:** Configuring the Solver Parameters dialog box